ICT 6535: Advanced Artificial Intelligence

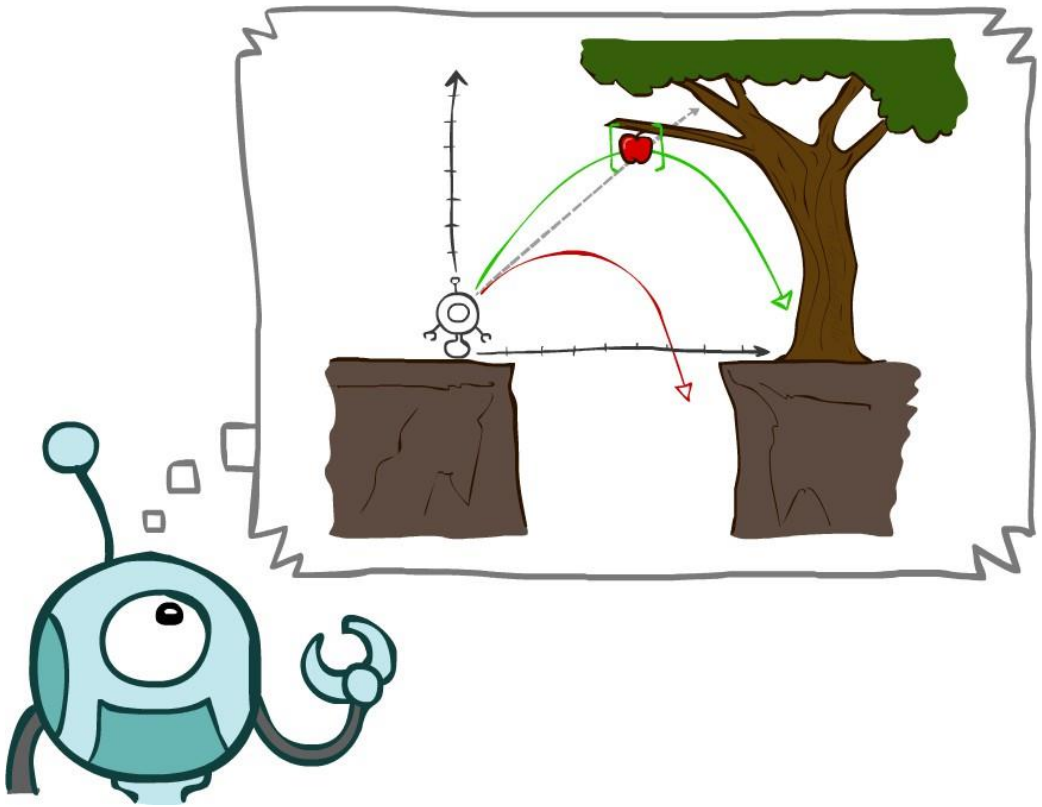
Lecture 3 — Uninformed Search in AI

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Bangladesh University of Engineering and Technology (BUET)

Contents

- **Uninformed Search Methods**
  - Breadth-First Search (BFS)
  - Uniform-Cost Search (UCS)
  - Depth-First Search (DFS)
  - Depth-Limited Search (DLS)
  - Iterative Deepening Search (IDS)
  - Bidirectional Search



Search Problems



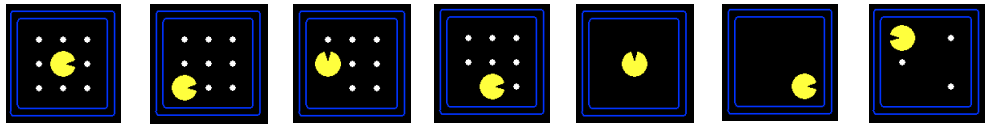
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Search Problems

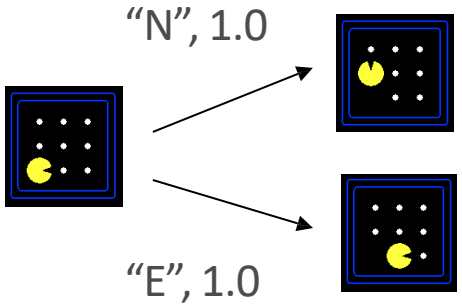
- ***Artificial Intelligence*** is the study of building agents that act rationally. Most of the time, these agents perform some kind of search algorithm in the background in order to achieve their Goal, in Goal based Agent system.
- A search problem consists of:
  - **A State Space:** Set of all possible states where an agent can be.
  - **A Start State:** The state from where the search begins.
  - **A successor function**
  - **A Goal Test:** A function that looks at the current state whether or not it is the goal state.
- The **Solution** to a search problem is a sequence of actions, called the **plan** that transforms the start state to the goal state.
- This plan is achieved through **search algorithms**.

Search Problems

- A search problem consists of:
  - A state space



  - A successor function (with actions, costs)

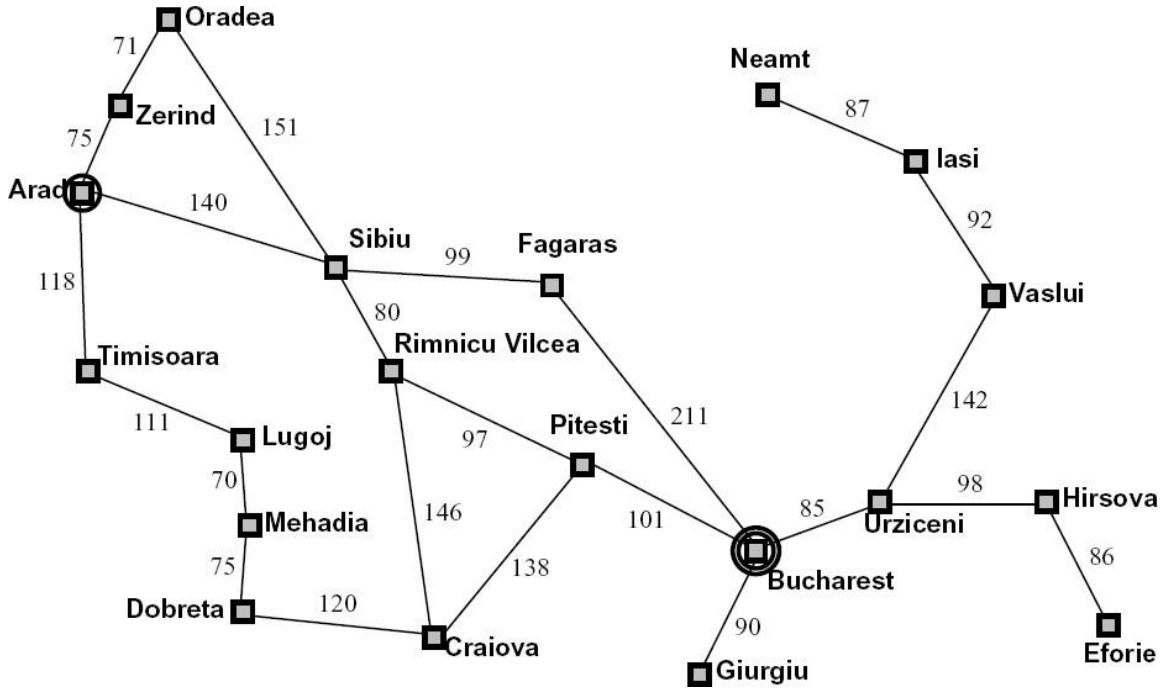


  - A start state and a goal test

- A solution is a sequence of actions (a plan) which transforms the start state to a goal state.

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Example: Traveling in Romania

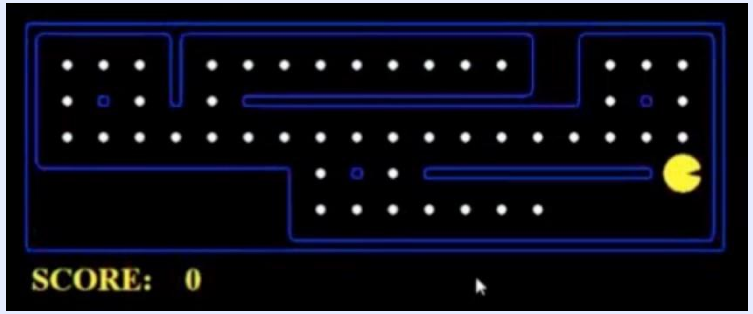


- State space:
  - Cities
- Successor function:
  - Roads: Go to adjacent city with cost = distance
- Start state:
  - Arad
- Goal test:
  - Is state == Bucharest?
- Slution?

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What's in a State Space?

The world state includes every last detail of the environment



A search state keeps only the details needed for planning (abstraction)

- Problem: Pathing
  - States: (x,y) location
  - Actions: NSEW
  - Successor: update location only
  - Goal test: is (x,y)=END

- Problem: Eat-All-Dots
  - States: {(x,y), dot booleans}
  - Actions: NSEW
  - Successor: update location and possibly a dot boolean
  - Goal test: dots all false

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State Space Sizes?

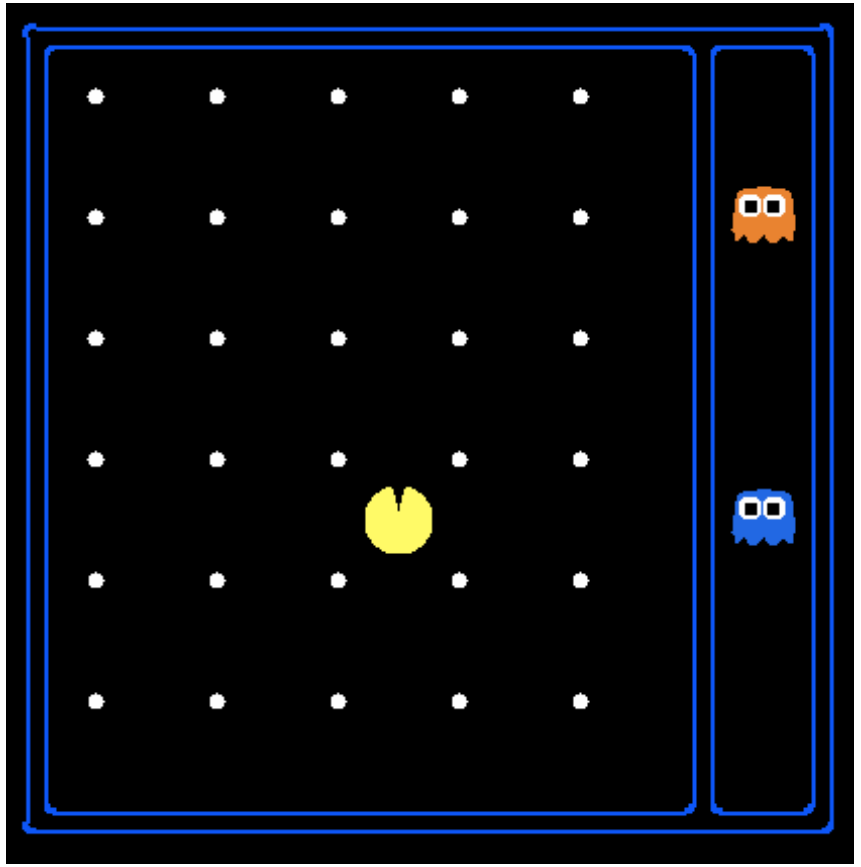
- World state:
  - Agent position on 12x10 grid
  - Status of 30 dots
  - Ghost positions on 12x1 grid
  - Agent facing matters: NSEW

- How many
  - World states?

    $120x(2^{30})x(12^2)x4$
  - States for pathing?

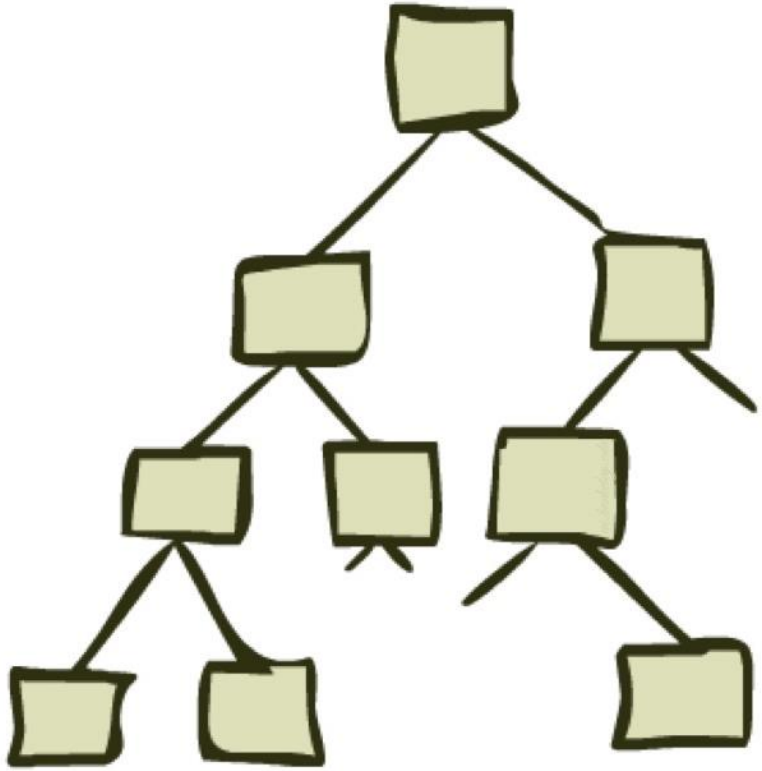
    120
  - States for eat-all-dots?

    $120x(2^{30})$



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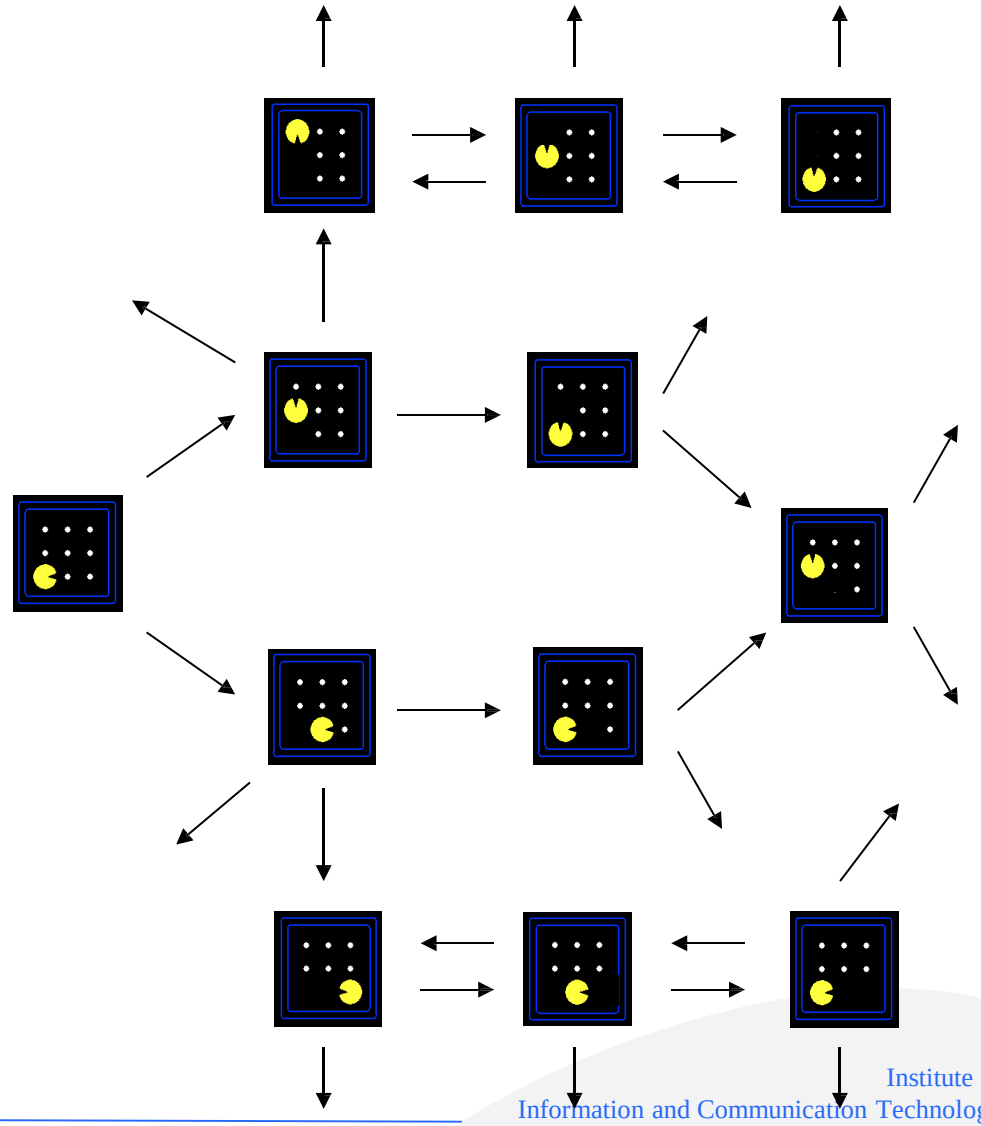
State Space Graphs and Search Trees



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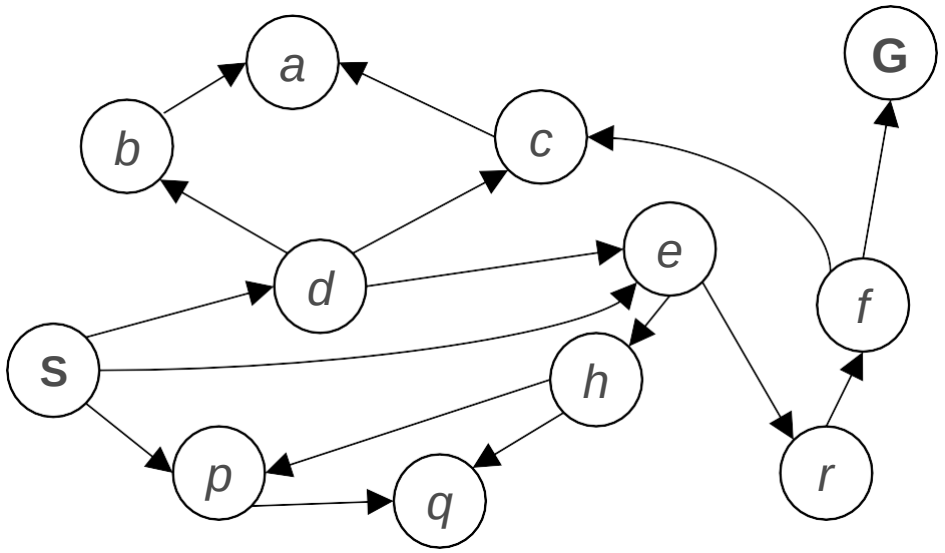
State Space Graphs

- State space graph: A mathematical representation of a search problem
  - Nodes are (abstracted) world configurations
  - Arcs represent successors (action results)
  - The goal test is a set of goal nodes (maybe only one)
- In a state space graph, each state occurs only once!
- We can rarely build this full graph in memory (it's too big), but it's a useful idea



State Space Graphs

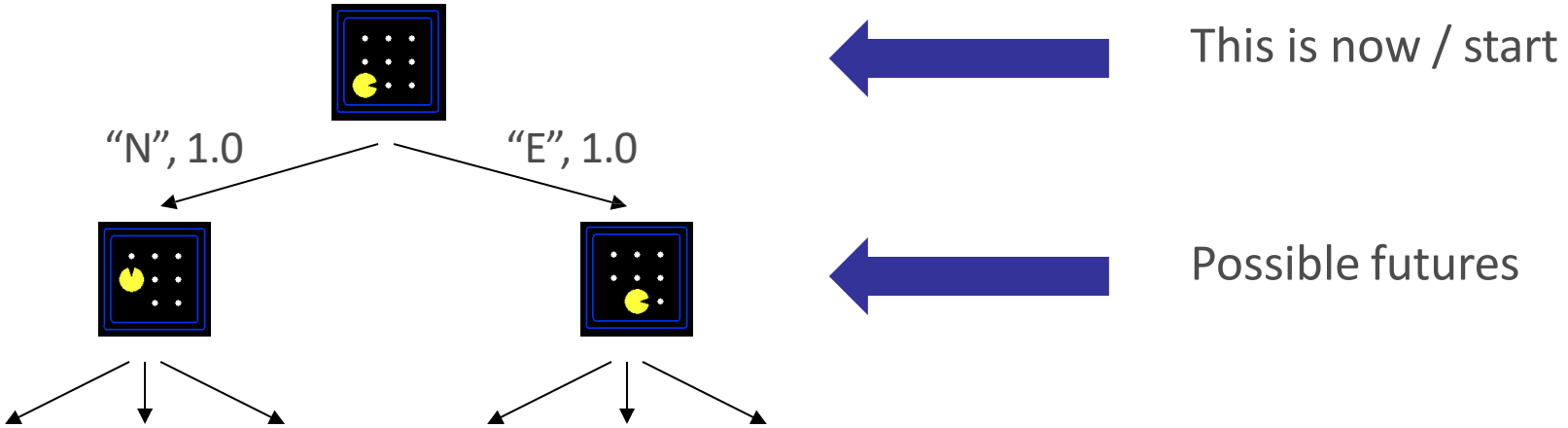
- State space graph: A mathematical representation of a search problem
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- We can rarely build this full graph in memory (it's too big), but it's a useful idea



Tiny state space graph for a tiny search problem

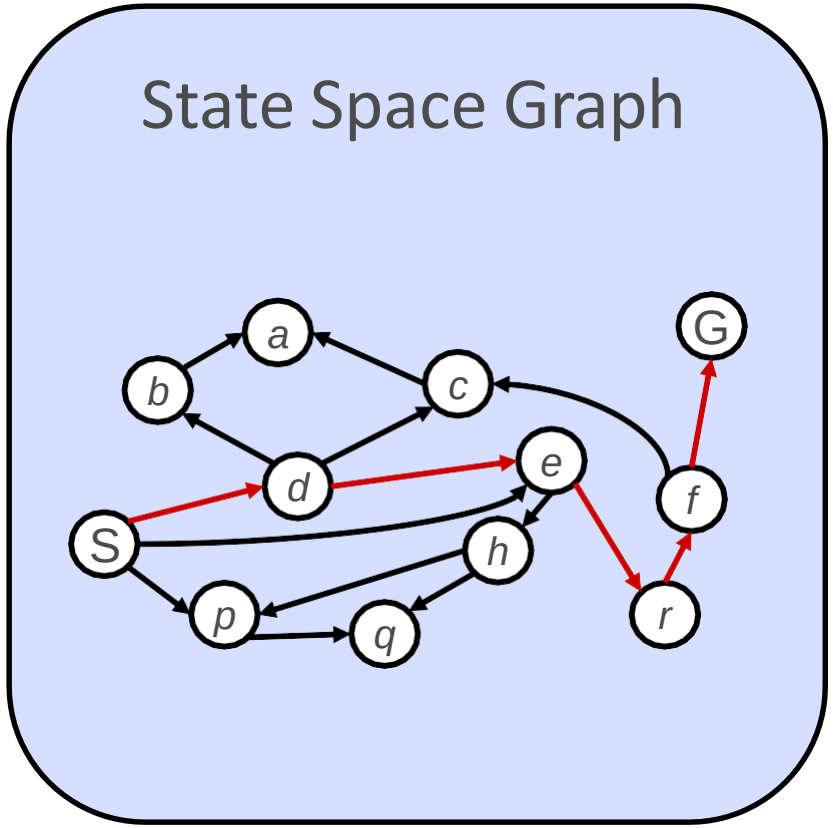
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Search Trees



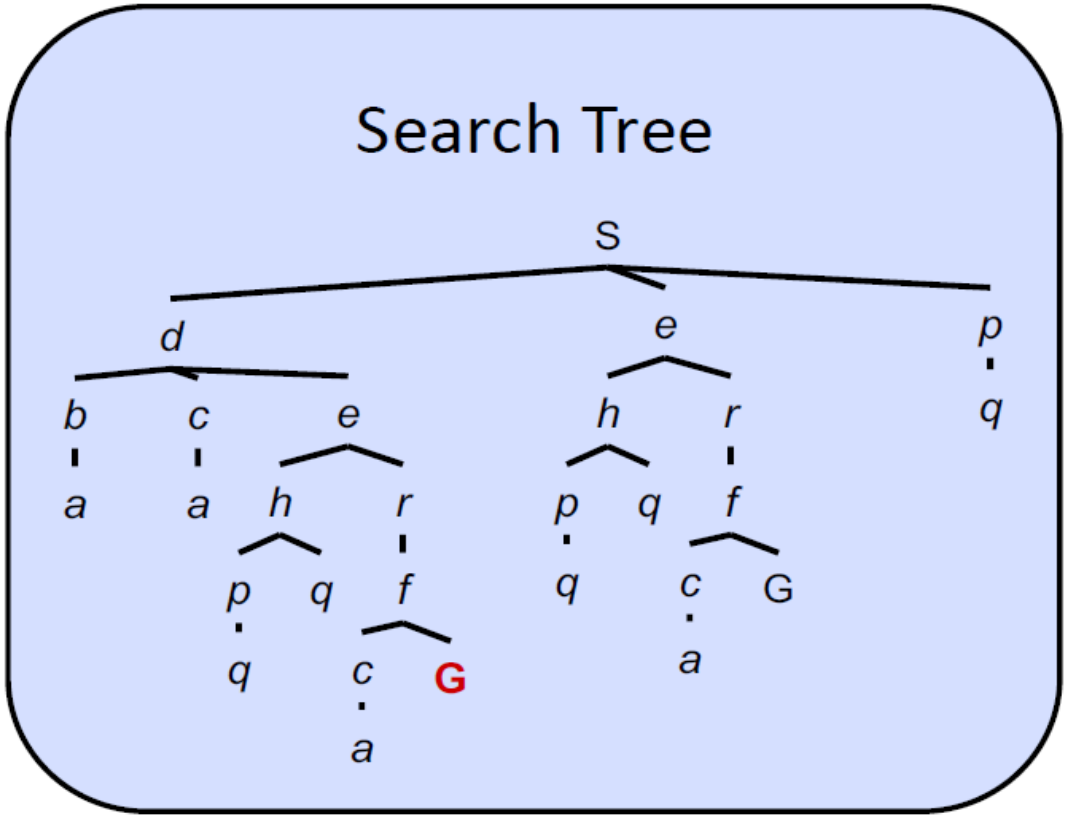
- A search tree:
  - A "what if" tree of plans and their outcomes
  - The start state is the root node
  - Children correspond to successors
  - Nodes show states, but correspond to PLANS that achieve those states
  - For most problems, we can never actually build the whole tree

State Space Graphs vs. Search Trees



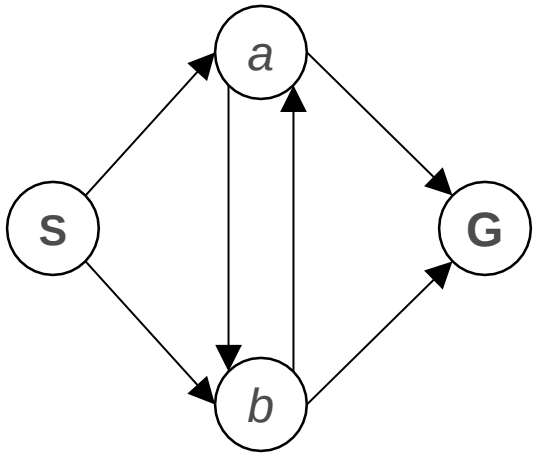
Each NODE in the search tree is an entire PATH in the state space graph.

We construct both on demand – and we construct as little as possible.



Quiz: State Space Graphs vs. Search Trees

Consider this 4-state graph:

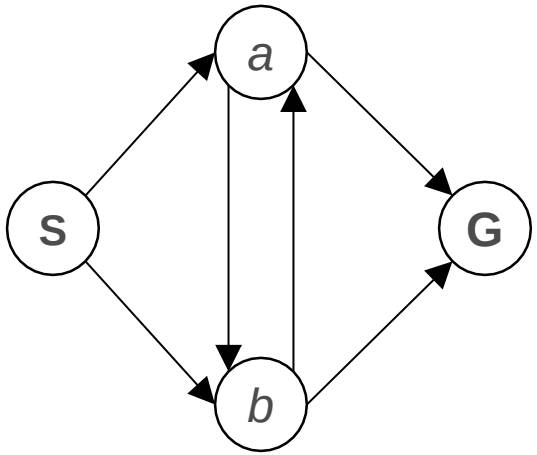


How big is its search tree (from S)?

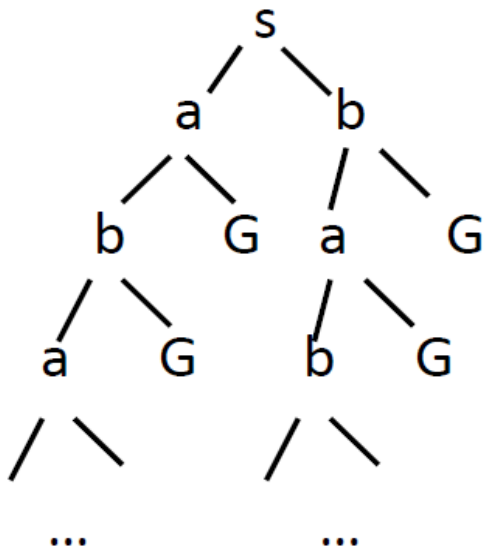


Quiz: State Space Graphs vs. Search Trees

Consider this 4-state graph:



How big is its search tree (from S)?

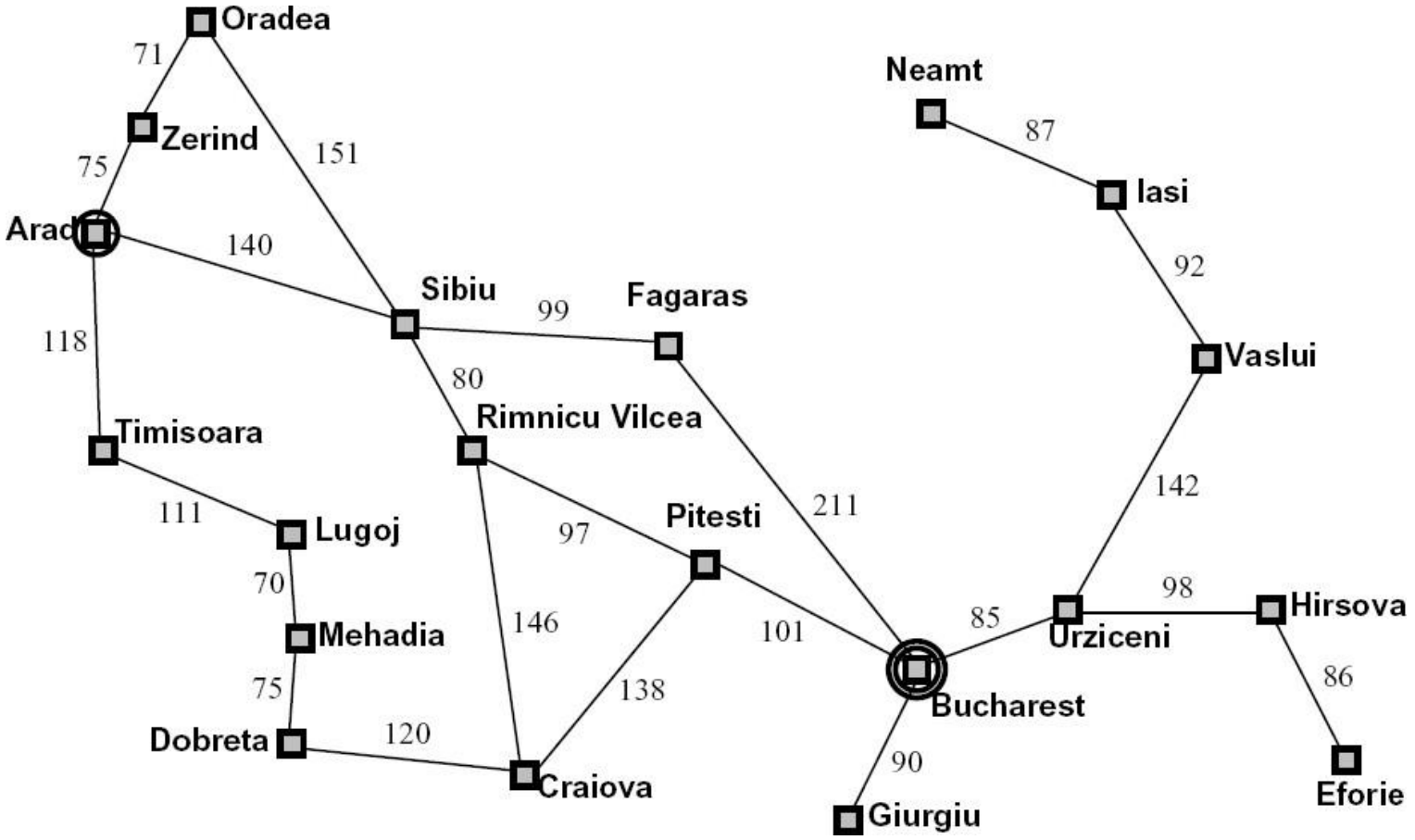




Important: Lots of repeated structure in the search tree!

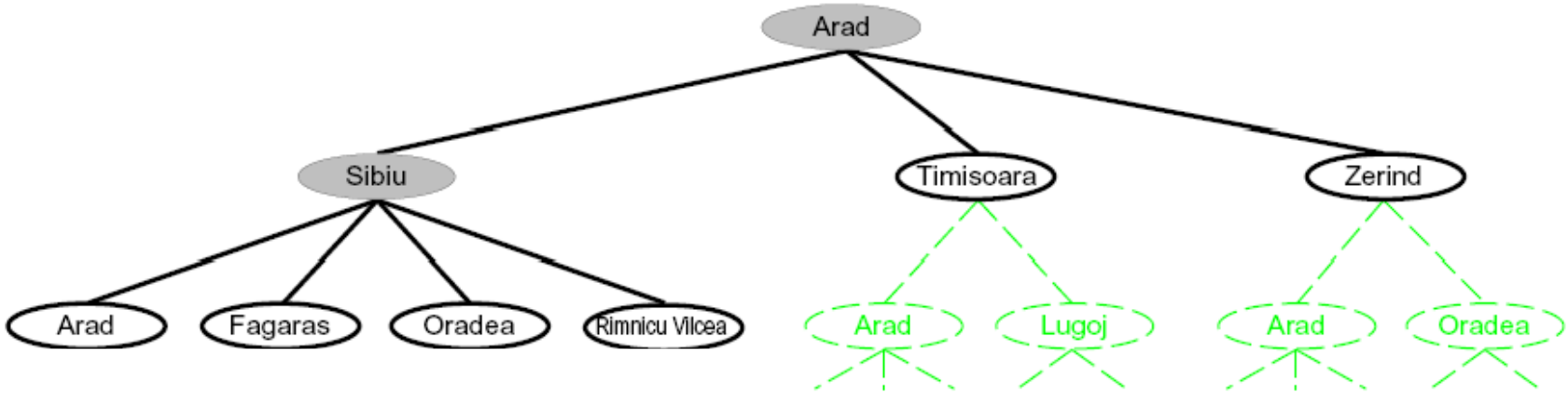
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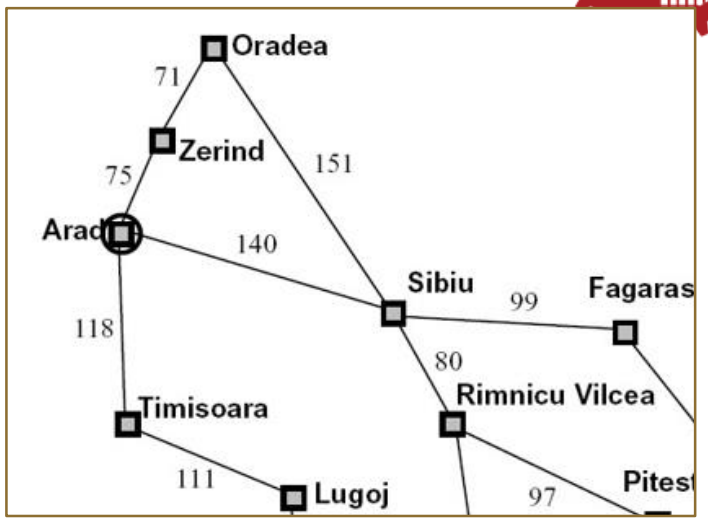
Search Example: Romania



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Searching with a Search Tree





- Search:
  - Expand out potential plans (tree nodes)
  - Maintain a fringe of partial plans under consideration
  - Try to expand as few tree nodes as possible

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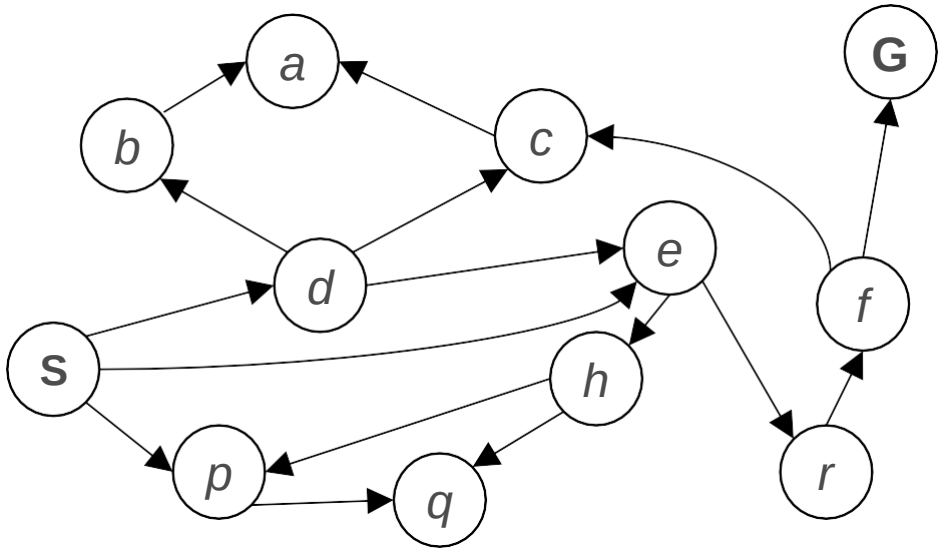
General Tree Search

```
function TREE-SEARCH( problem, strategy) returns a solution, or failure
    initialize the search tree using the initial state of problem
    loop do
        if there are no candidates for expansion then return failure
        choose a leaf node for expansion according to strategy
        if the node contains a goal state then return the corresponding solution
        else expand the node and add the resulting nodes to the search tree
    end
```

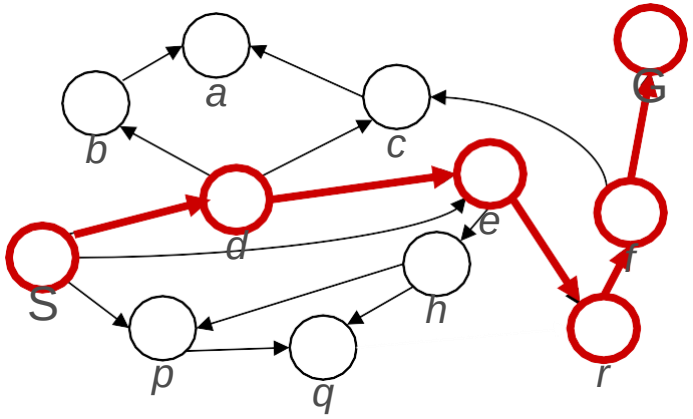
- Important ideas:
  - Fringe
  - Expansion
  - Exploration strategy
- Main question: which fringe nodes to explore?

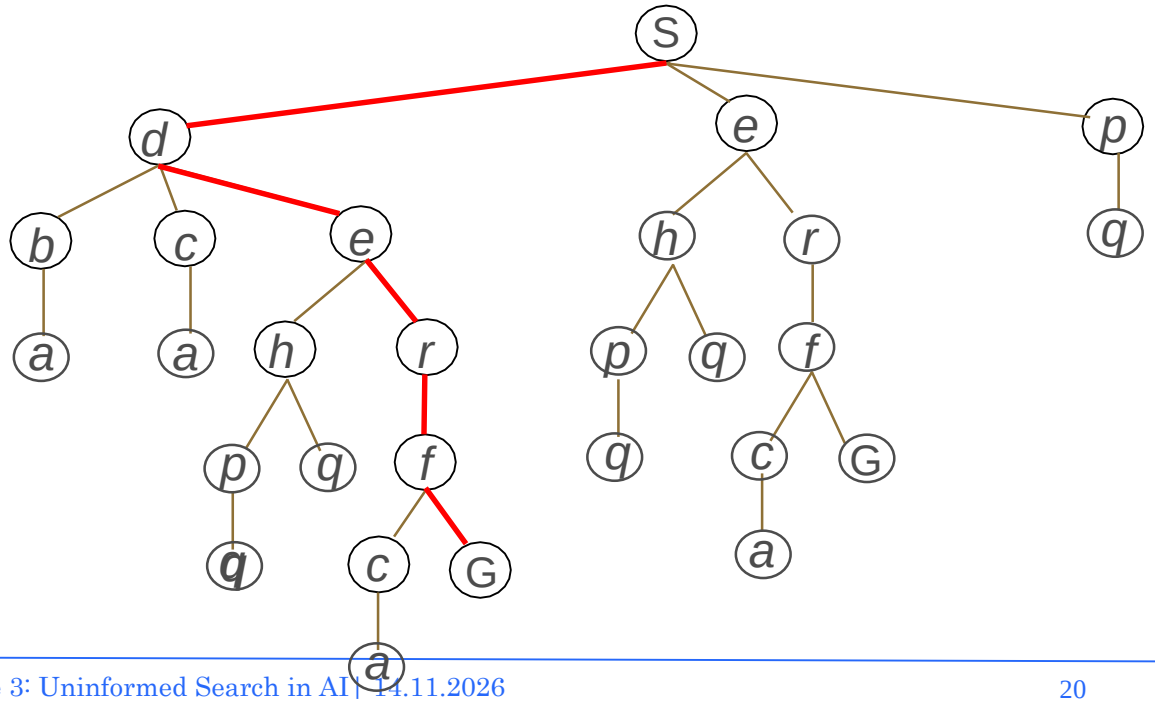
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Example: Tree Search



Example: Tree Search





~~S~~

~~S → d~~

S → e

S → p

S → d → b

S → d → c

~~S → d → e~~

S → d → e → h

~~S → d → e → r~~

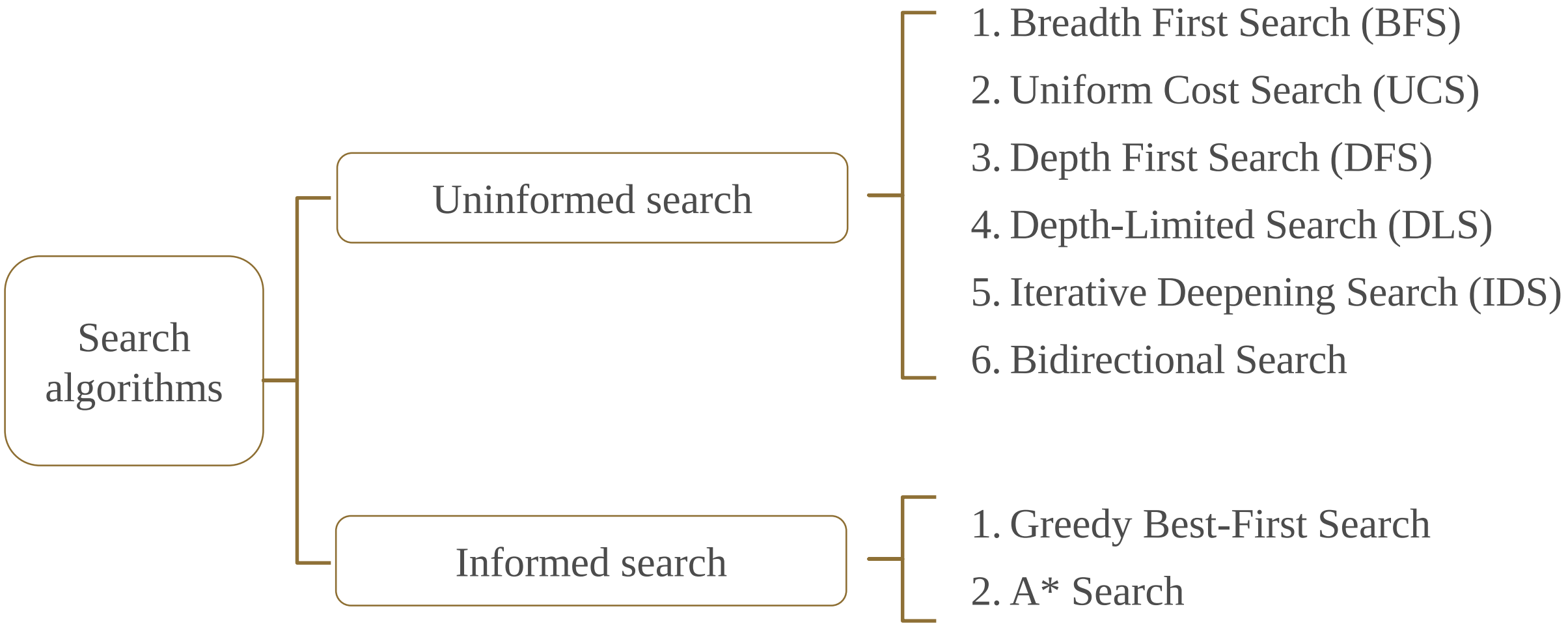
~~S → d → e → r → f~~

S → d → e → r → f → c

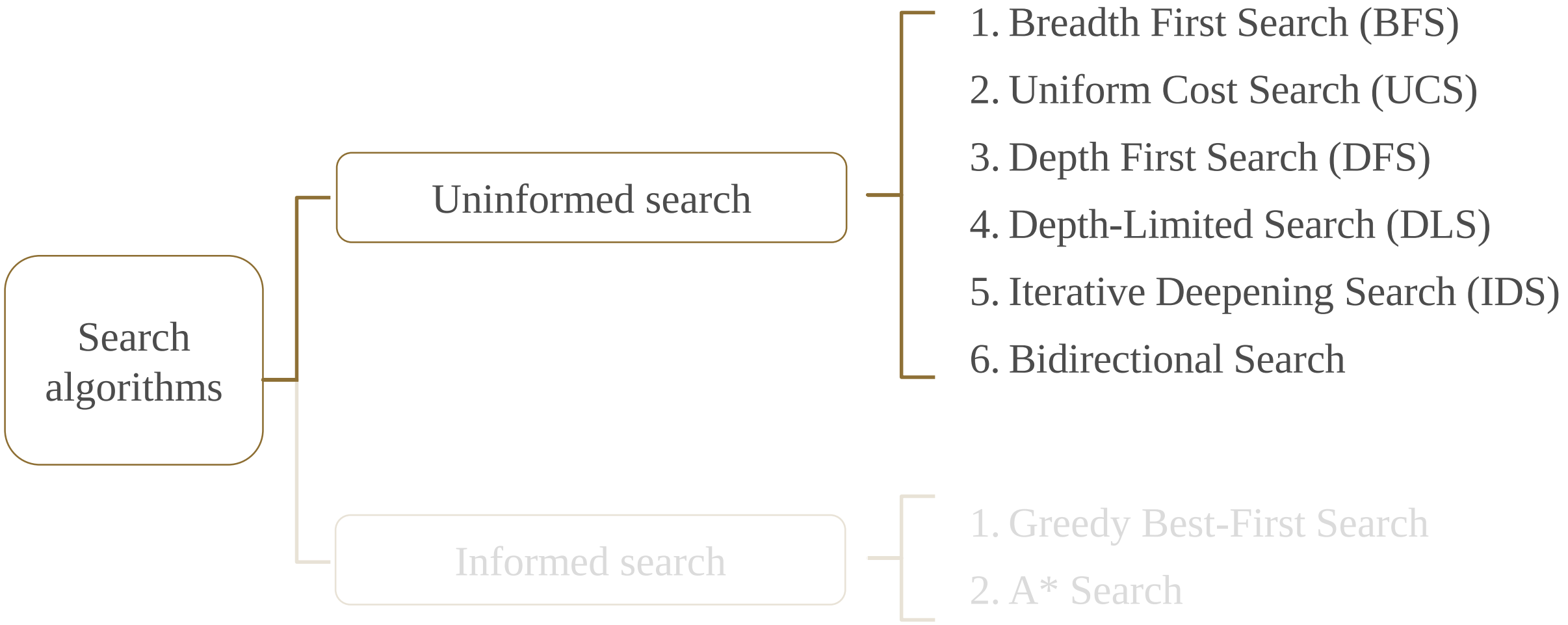
~~S → d → e → r → f → G~~

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Types of Search Algorithms



Uninformed Search Algorithms



Search Algorithm Properties



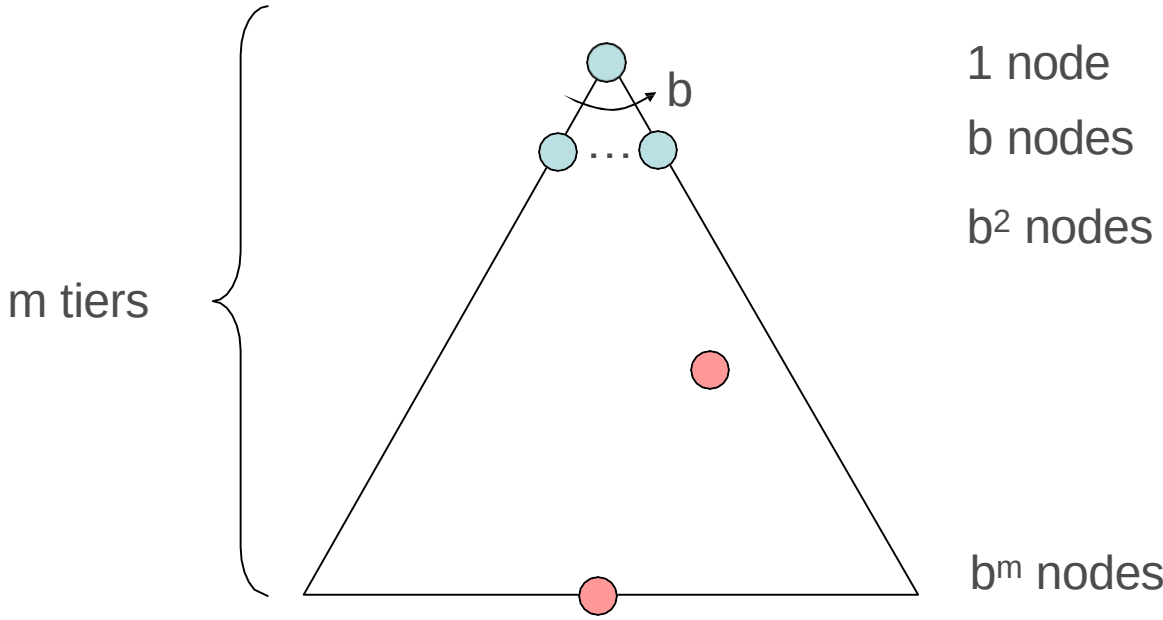
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Search Algorithm Properties

- Complete: Guaranteed to find a solution if one exists?
- Optimal: Guaranteed to find the least cost path?
- Time complexity?
- Space complexity?

- Cartoon of search tree:
  - *b* is the branching factor
  - *m* is the maximum depth
  - solutions at various depths

- Number of nodes in entire tree?
  - $1 + b + b^2 + ... b^m = O(b^m)$

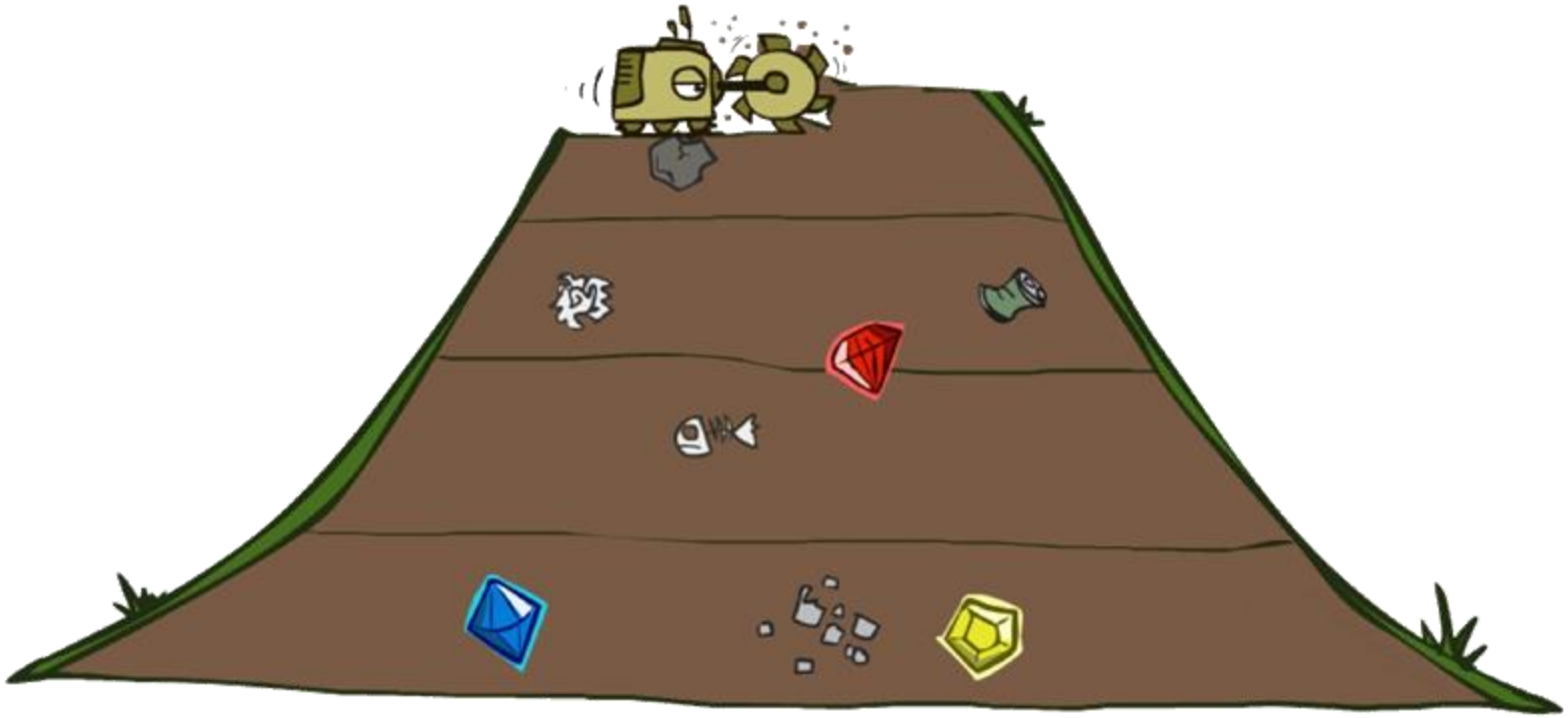


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Uninformed Search Algorithms

- The uninformed search algorithms have no additional information on the goal node other than the one provided in the problem definition.
- The plans to reach the goal state from the start state differ only by the order and/or length of actions.
- Uninformed search is also called **Blind** search.
- These algorithms can only generate the successors and differentiate between the goal state and non goal state.
- All search strategies are distinguished by the order in which nodes are expanded.

Breadth-First Search (BFS)



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Breadth-First Search (BFS)

- **Breadth-first search** is a simple strategy in which the root node is expanded first, then all the successors of the root node are expanded next, then their successors, and so on.
  - Breadth-first search is an instance of the general graph-search algorithm in which the shallowest unexpanded node is chosen for expansion.
  - This is achieved very simply by using a FIFO queue for the frontier.
    - Thus, new nodes (which are always deeper than their parents) go to the back of the queue, and old nodes, which are shallower than the new nodes, get expanded first.
- The goal test is applied to each node when it is generated rather than when it is selected for expansion.

Breadth-First Search (BFS)

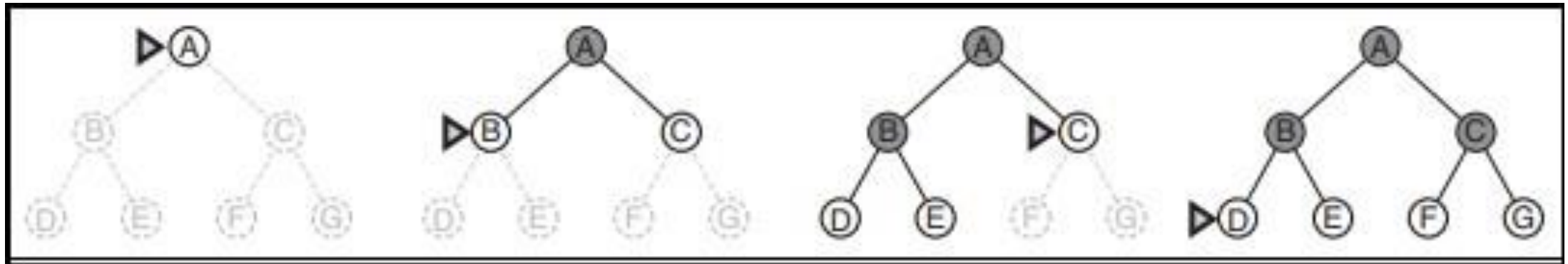


Figure 3.12 Breadth-first search on a simple binary tree. At each stage, the node to be expanded next is indicated by a marker.

BFS is complete

- if the shallowest goal node is at some finite depth *d*, breadth-first search will eventually find it after generating all shallower nodes,
- provided the branching factor *b* is finite.

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Breadth-First Search (BFS)

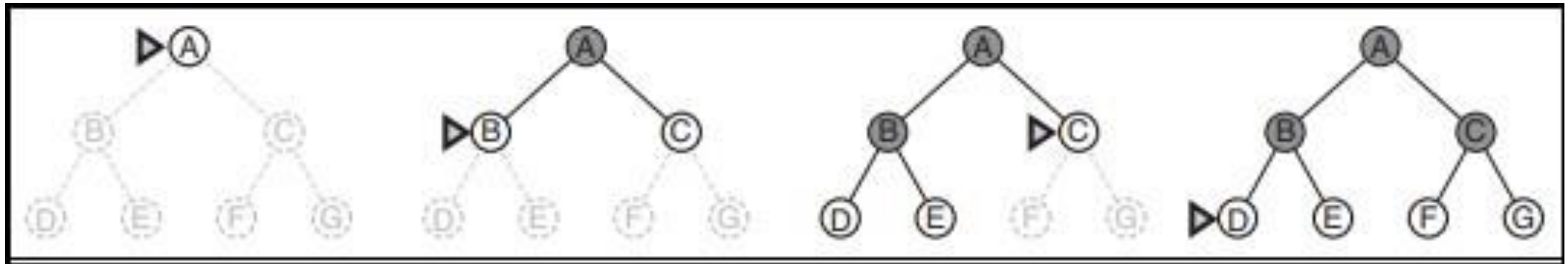


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BFS is complete

- if the shallowest goal node is at some finite depth *d*, breadth-first search will eventually find it after generating all shallower nodes
- provided the branching factor *b* is finite

As soon as a goal node is generated, we know it is the shallowest goal node because all shallower nodes must have been generated already and failed the goal test.

Breadth-First Search (BFS)

Time Complexity

- Imagine searching a uniform tree where every state has b successors.
- The root of the search tree generates b nodes at the first level, each of which generates b more nodes, for a total of b^2 at the second level.
- Each of these generates b more nodes, yielding b^3 nodes at the third level, and so on.
- Now suppose that the solution is at depth d. In the worst case, it is the last node generated at that level. Then the total number of nodes generated is

$$b + b^2 + b^3 + \cdots + b^d = O(b^d)$$

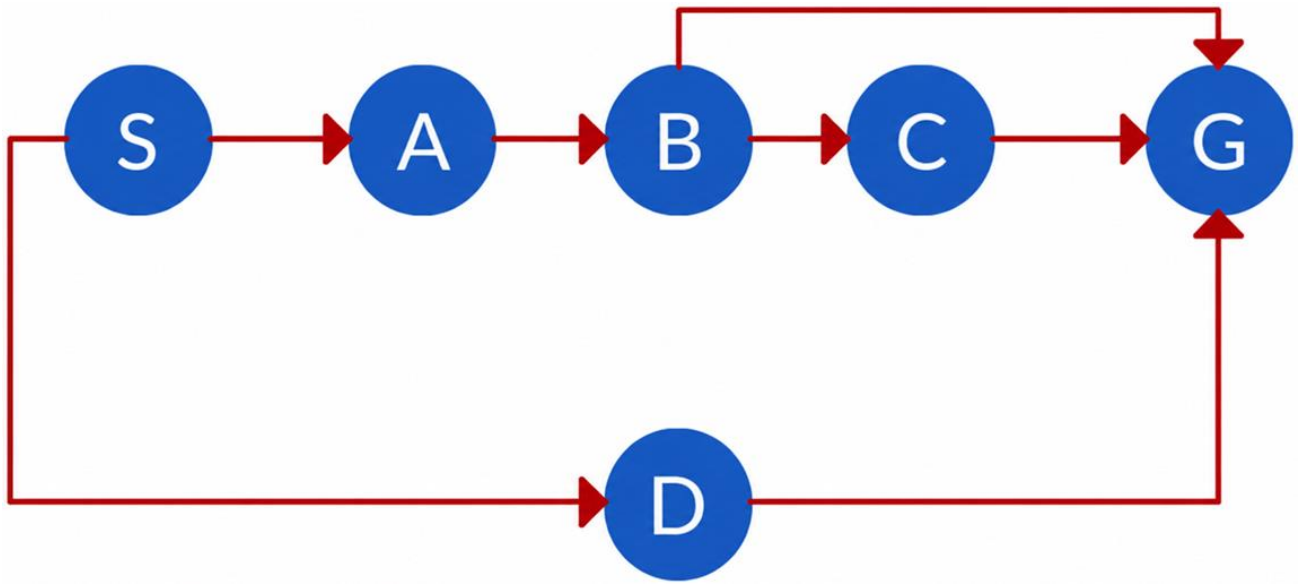
Space Complexity

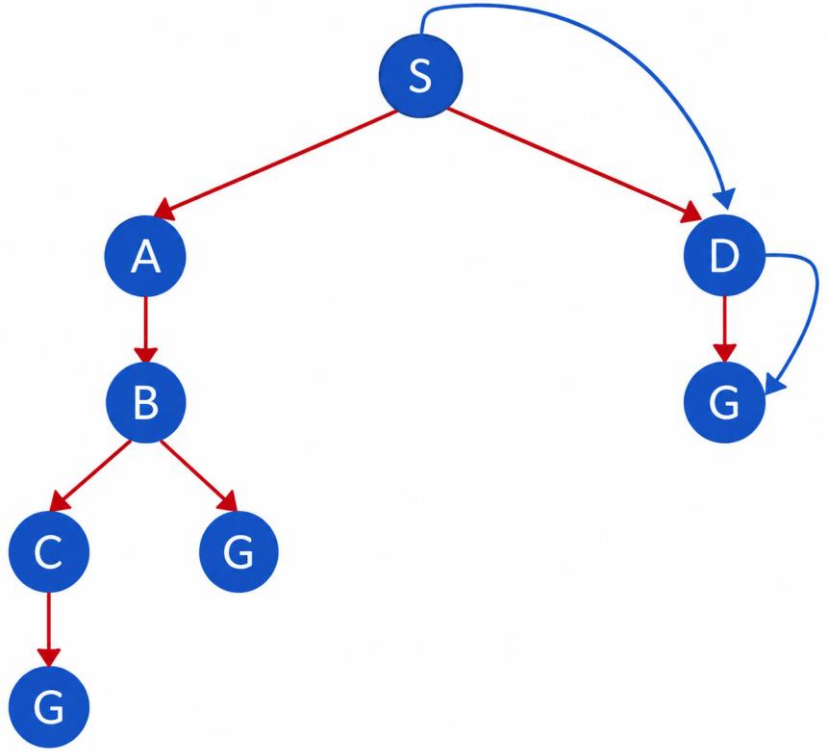
- every node generated remains in memory. There will be $O(b^{d-1})$ nodes in the explored set and $O(b^d)$ nodes in the frontier, so the space complexity is $O(b^d)$.

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Breadth-First Search (BFS)

Which solution would BFS find to move from node *S* to node *G* if run on the graph below?



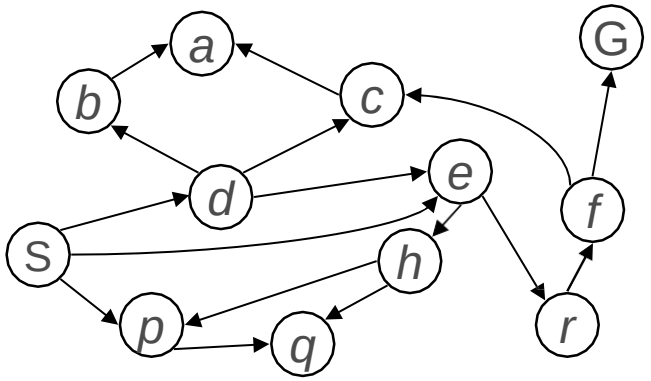


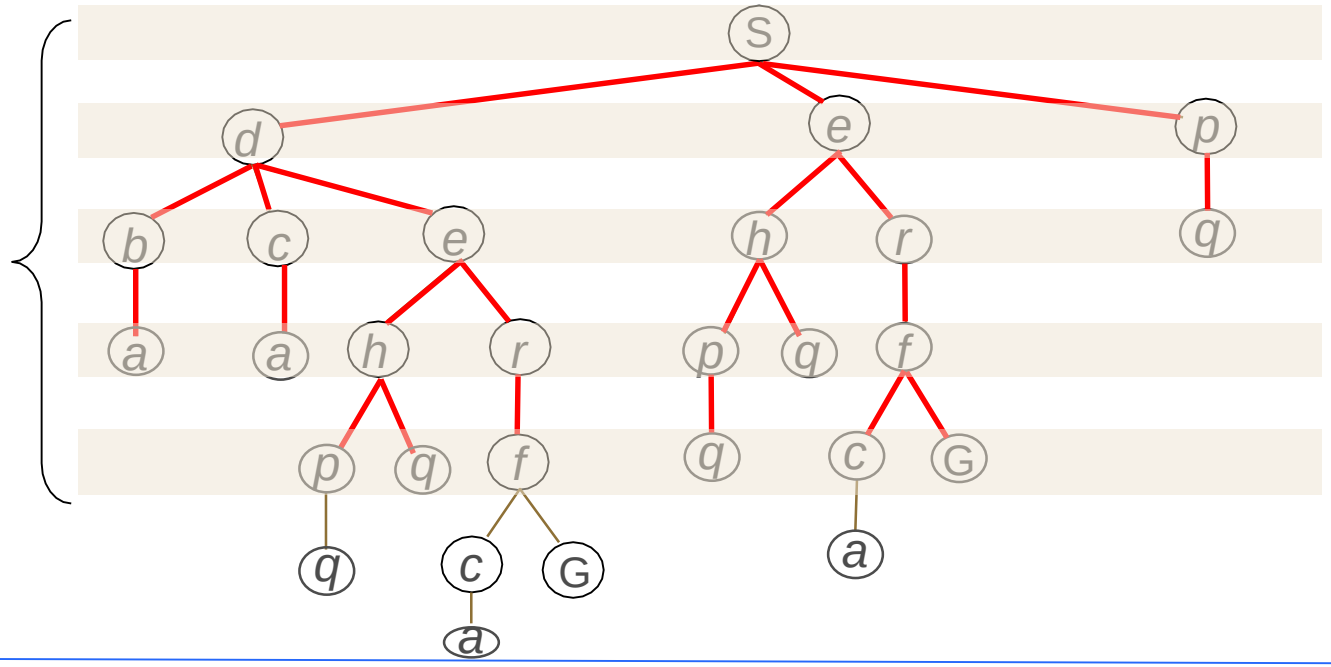
Path: S → D → G

Breadth-First Search (BFS)

Strategy: expand a shallowest node first

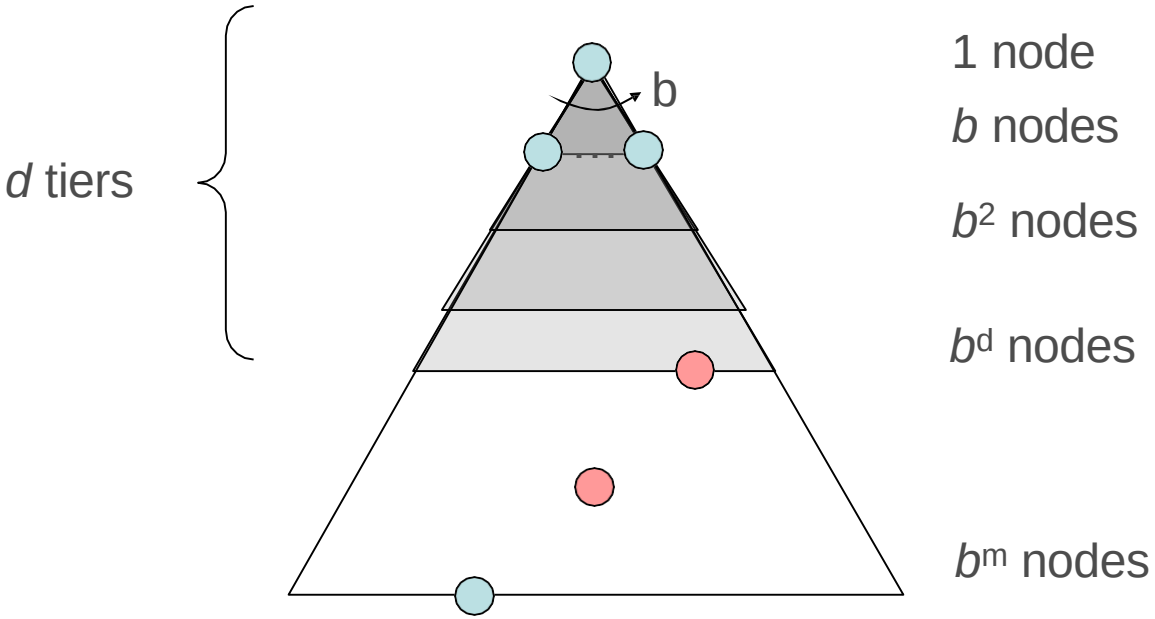
Implementation: Fringe is a FIFO queue





Breadth-First Search (BFS) Properties

- What nodes does BFS expand?
  - Processes all nodes above shallowest solution
  - Let depth of shallowest solution be *d*
  - Search takes time $O(b^d)$
- How much space does the fringe take?
  - At end it has roughly the last tier, so $O(b^s)$
- Is it complete?
  - *d* must be finite if a solution exists, so yes!
- Is it optimal?
  - Only if costs are all 1 (more on costs later)



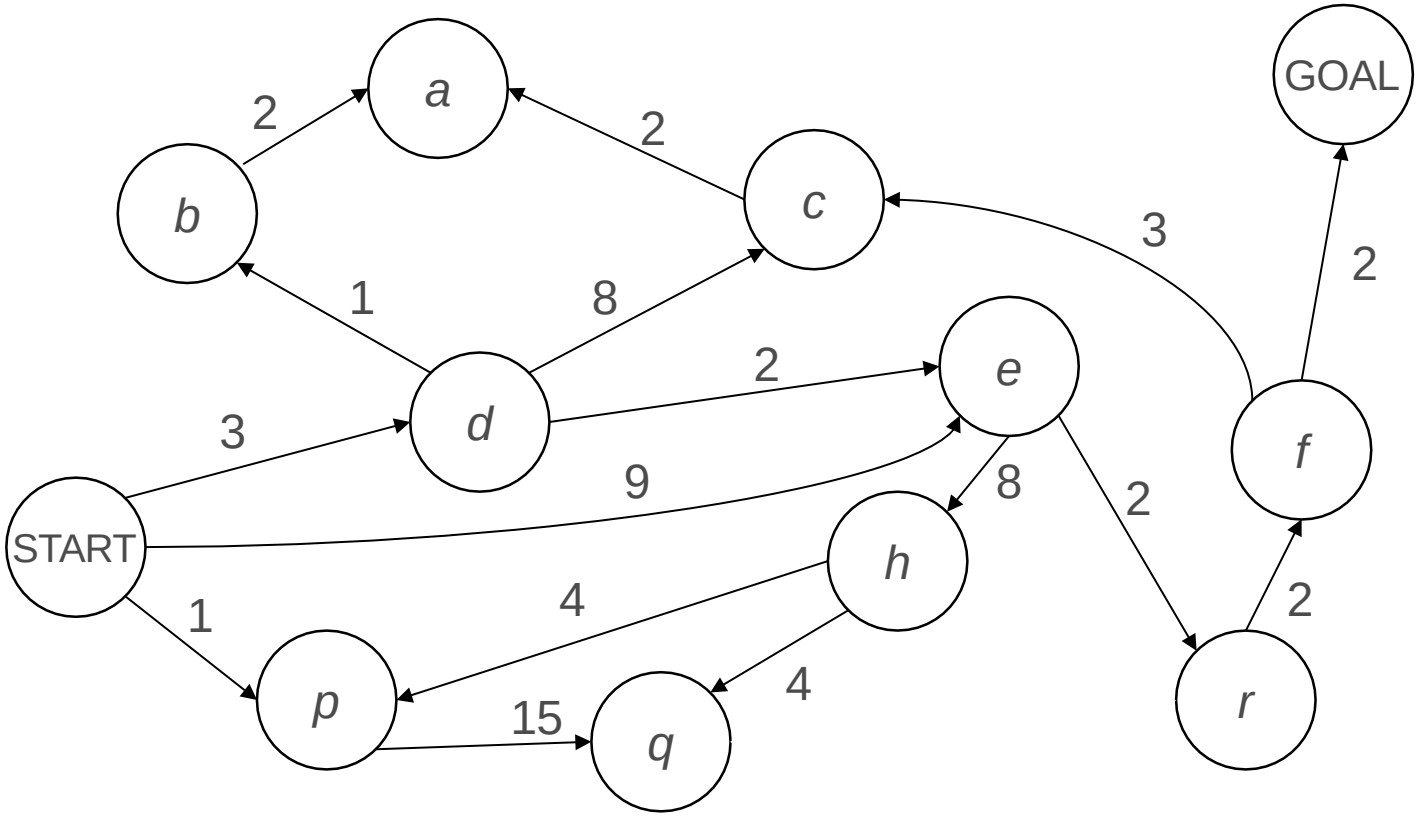
Breadth-First Search (BFS) Properties

Depth	Nodes	Time	Memory
2	110	.11 milliseconds	107 kilobytes
4	11,110	11 milliseconds	10.6 megabytes
6	10^6	1.1 seconds	1 gigabyte
8	10^8	2 minutes	103 gigabytes
10	10^{10}	3 hours	10 terabytes
12	10^{12}	13 days	1 petabyte
14	10^{14}	3.5 years	99 petabytes
16	10^{16}	350 years	10 exabytes

Figure 3.13 Time and memory requirements for breadth-first search. The numbers shown assume branching factor $b = 10$; 1 million nodes/second; 1000 bytes/node.

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Cost-Sensitive Search

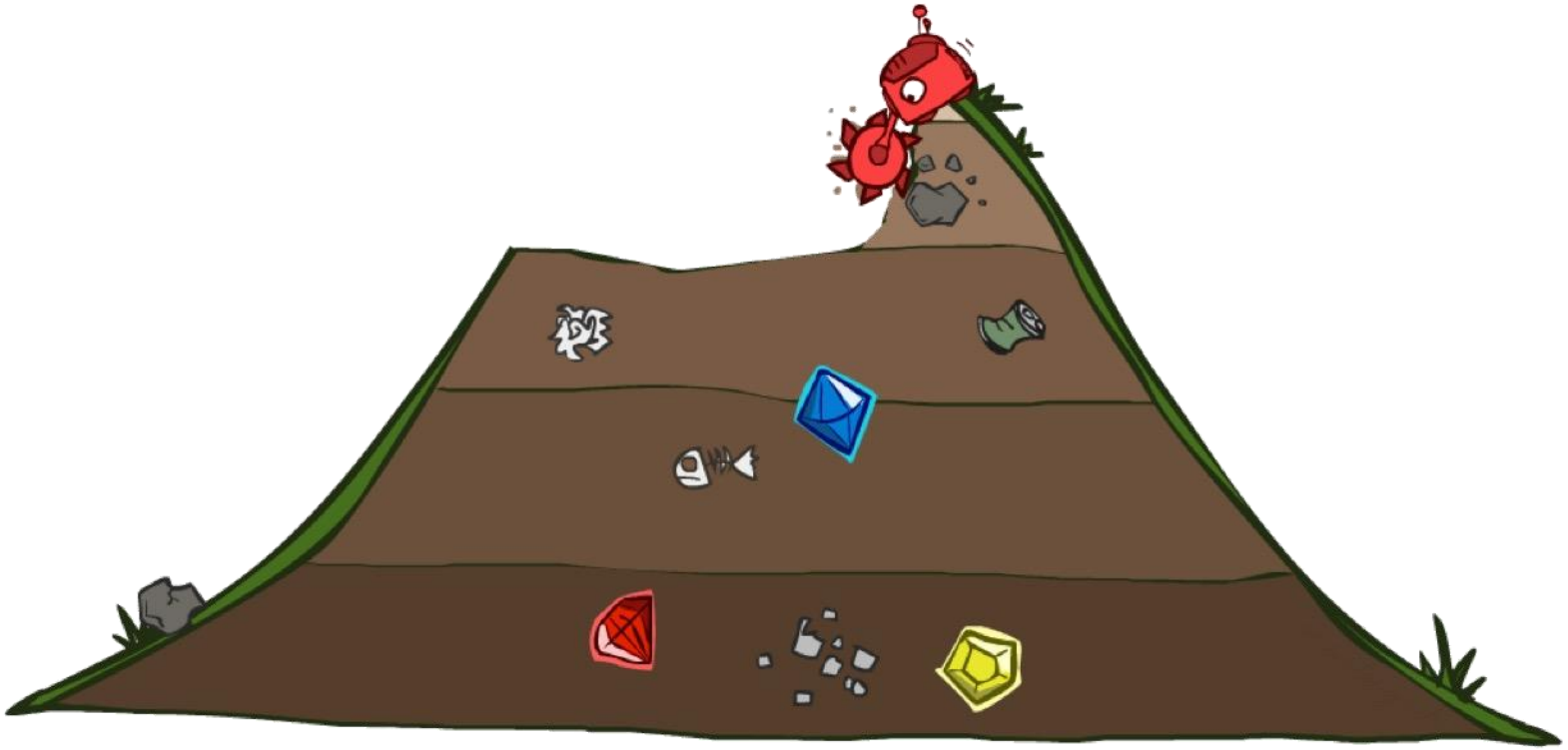


BFS finds the shortest path in terms of number of actions.

- It does not find the least-cost path.

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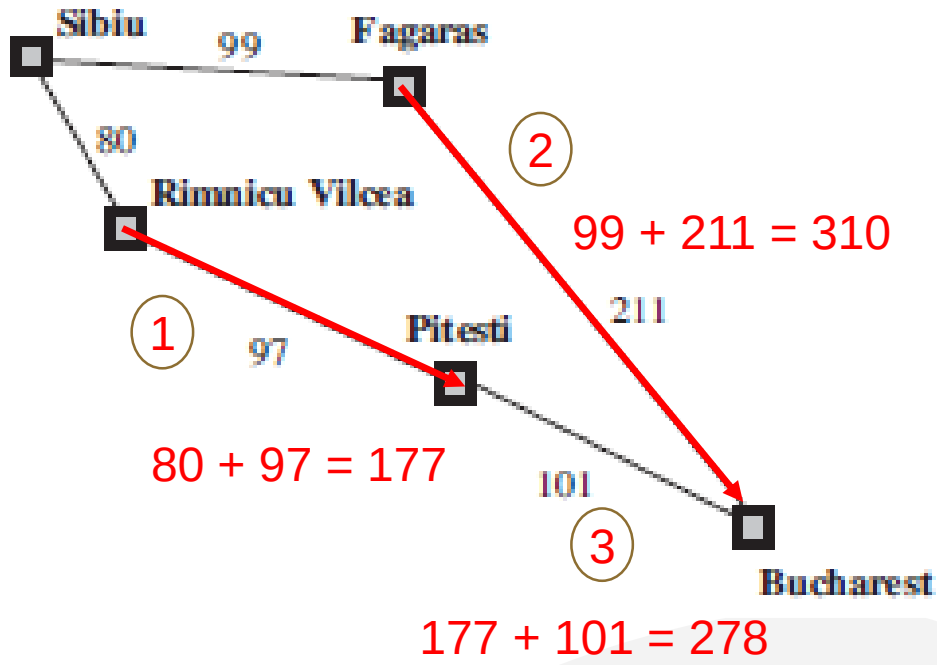
Uniform Cost Search (UCS)



Uniform Cost Search (UCS)

- While BFS is optimal only when all step costs are equal, but UCS is optimal for any step costs.
- In **Uniform cost search** method node with the *lowest path cost g(n)* is expanded rather the shallowest node.
- The goal test is applied to each node when it is selected for expansion rather than when it is generated.

  - Among two neighbors of Sibiu, Rimnicu Vilcea is expanded first, because of it is the least-cost node.
  - It adds Pitesti to a total path cost of 177 after step (1).
  - After step (1), Fagaras is the least-cost node and thus expanded, adding Bucharest to a cost of 310 in step (2).
  - Although the goal node 'Bucharest' is generated, UCS keeps searching and expands the current least-cost node 'Pitesti', reaching Bucharest with a new total cost of 278.

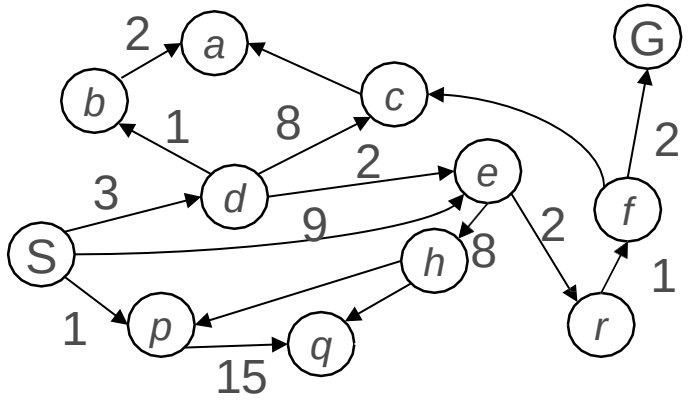


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Uniform Cost Search (UCS)

Strategy: expand a cheapest node first.

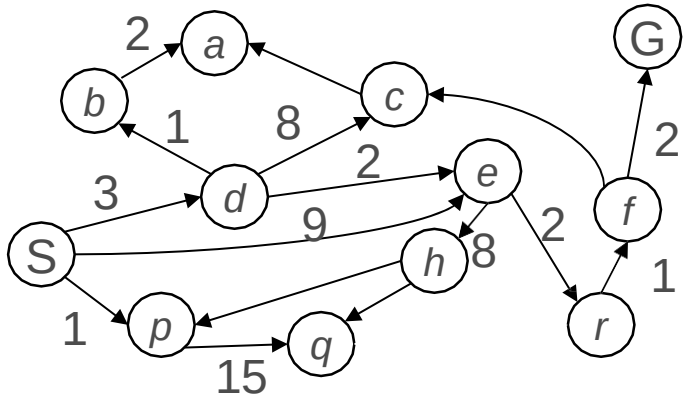
Fringe is a priority queue (priority: cumulative cost; numbers in tree refer to cumulative cost)

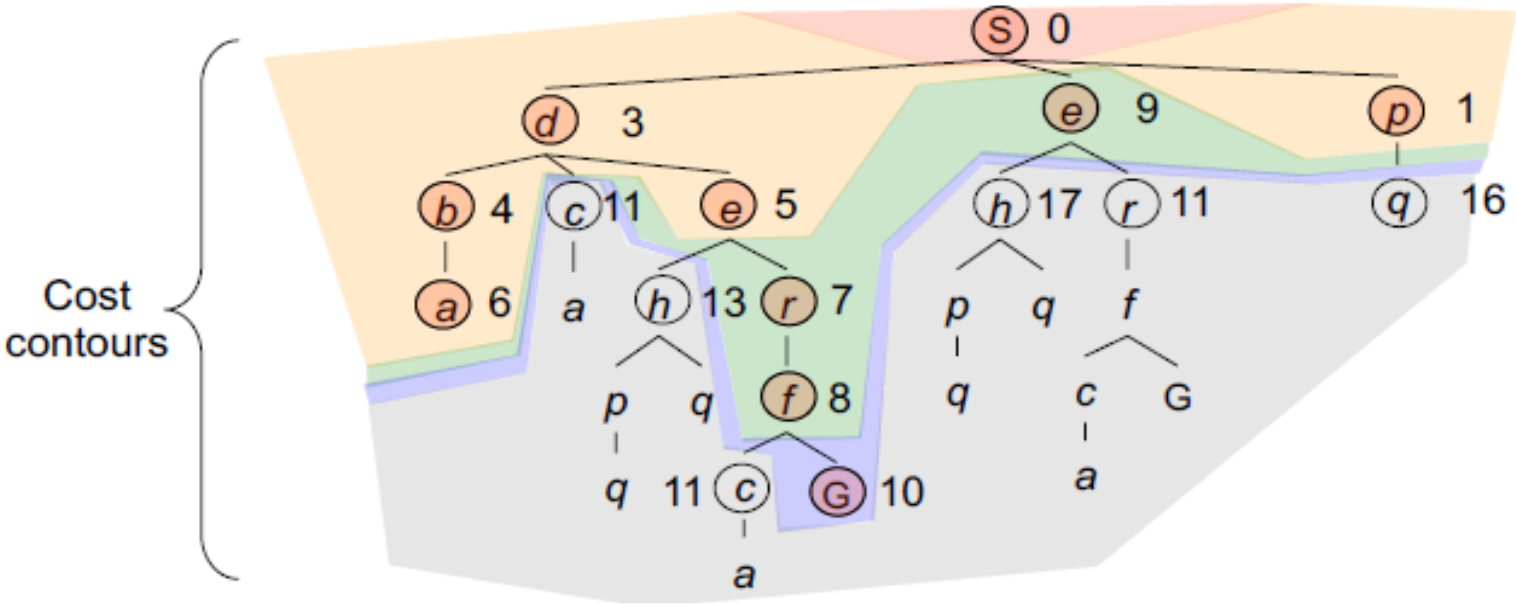


Uniform Cost Search (UCS)

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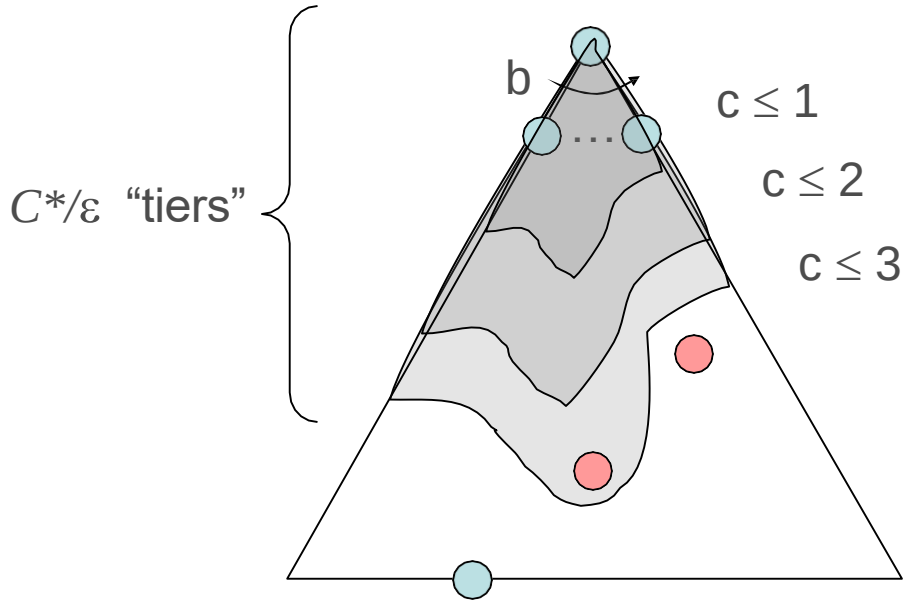




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Uniform Cost Search (UCS) Properties

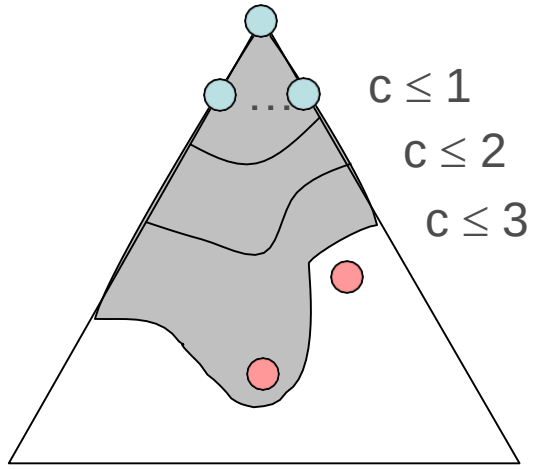
- What nodes does UCS expand?
  - Processes all nodes with cost less than cheapest solution!
  - If that solution costs C^* and arcs cost at least ε, then the "effective depth" is roughly C^*/ε
  - Takes time $O(b^{1+[C^*/\varepsilon]})$ (exponential in effective depth)
- How much space does the fringe take?
  - Has roughly the last tier, so $O(b^{1+[C^*/\varepsilon]})$
- Is it complete?
  - Assuming best solution has a finite cost and minimum arc cost is positive, yes!
- Is it optimal?
  - Yes! (Proof next lecture via A*)



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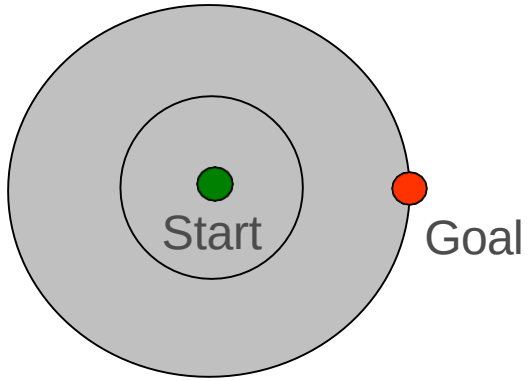
Uniform Cost Issues

- Remember: UCS explores increasing cost contours



- The good: UCS is complete and optimal!

- The bad:
  - Explores options in every “direction”
  - No information about goal location



- We’ll fix that soon!

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Depth-First Search (DFS)



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Depth-First Search (DFS)

- Depth-first search always expands the deepest node in the current frontier of the search tree.
- The search proceeds immediately to the deepest level of the search tree, where the nodes have no successors.
- As those nodes are expanded, they are dropped from the frontier, so then the search "backs up" to the next deepest node that still has unexplored successors
- Depth-first search uses a LIFO queue. A LIFO queue means that the most recently generated node is chosen for expansion.



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Depth-First Search (DFS)

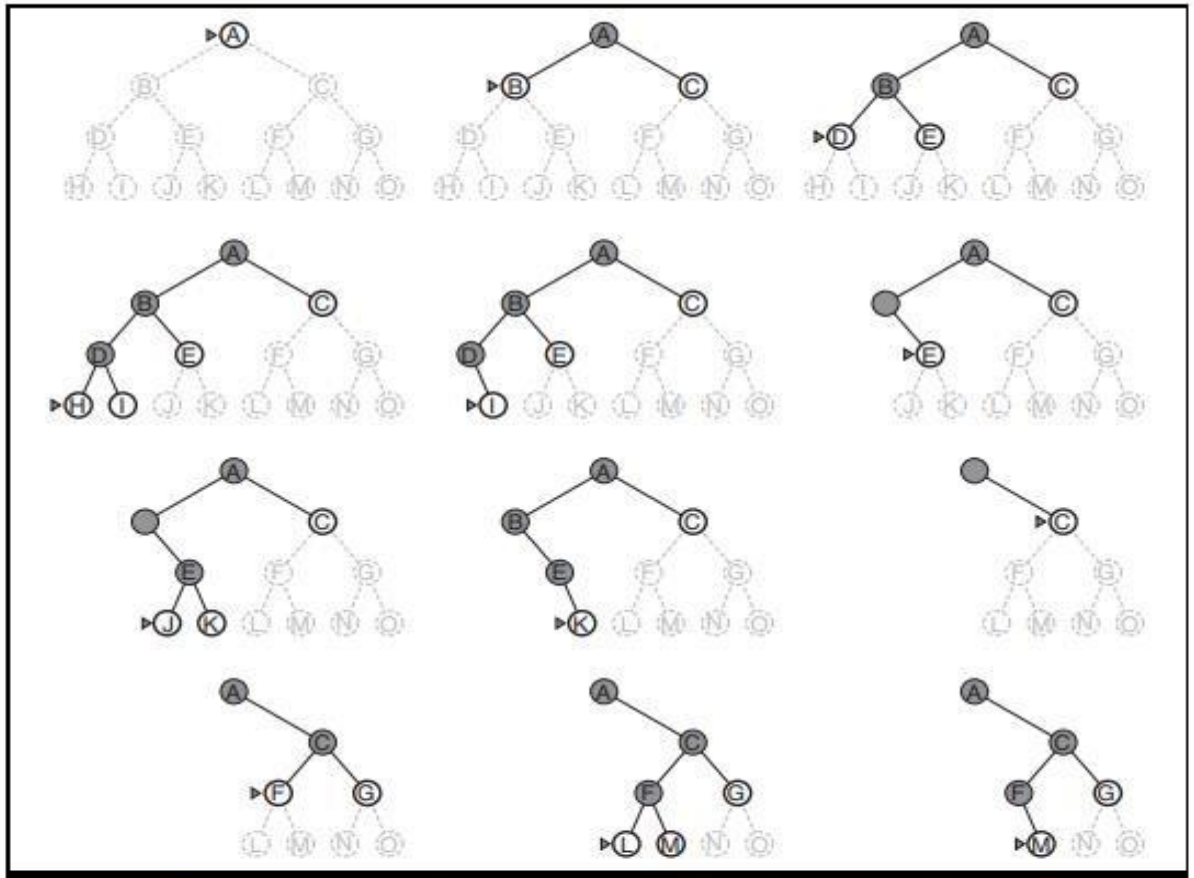
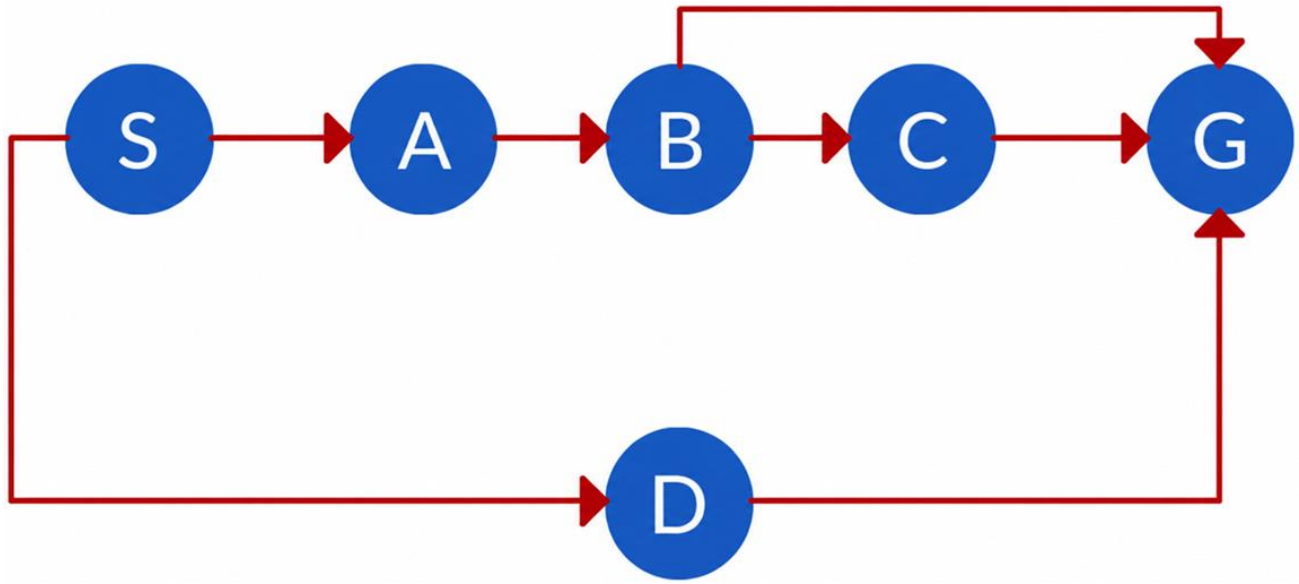


Figure 3.16 Depth-first search on a binary tree. The unexplored region is shown in light gray. Explored nodes with no descendants in the frontier are removed from memory. Nodes at depth 3 have no successors and M is the only goal node.

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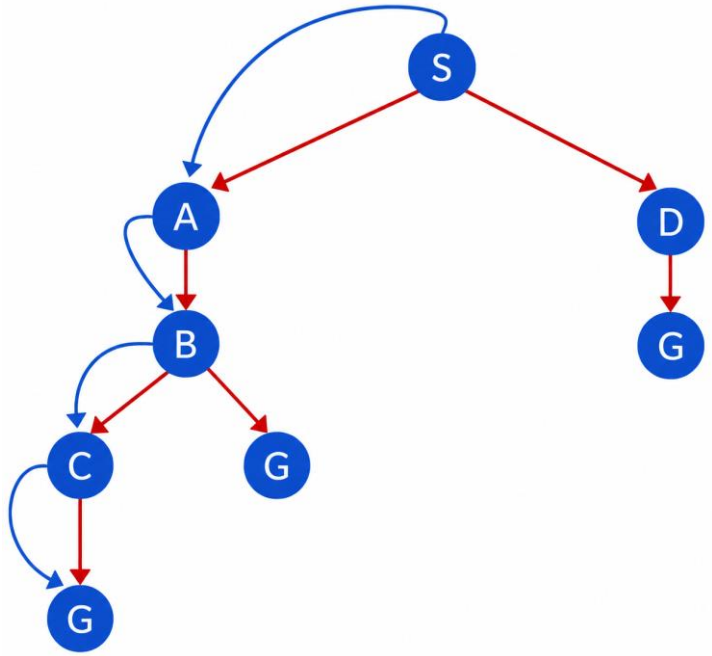
Depth-First Search (DFS)

Which solution would DFS find to move from node *S* to node *G* if run on the graph below?



Depth-First Search (DFS)

- As DFS traverses the tree “deepest node first”, it would always pick the deeper branch until it reaches the solution (or it runs out of nodes, and goes to the next branch). The traversal is shown in blue arrows.



Path: S → A → B → C → G

Depth-First Search (DFS)

Time Complexity

- A depth-first tree search may generate all of the $O(b^m)$ nodes in the search tree, where m is the maximum depth of any node.
- A depth-first tree search needs to store only a single path from the root to a leaf node, along with the remaining unexpanded sibling nodes for each node on the path.

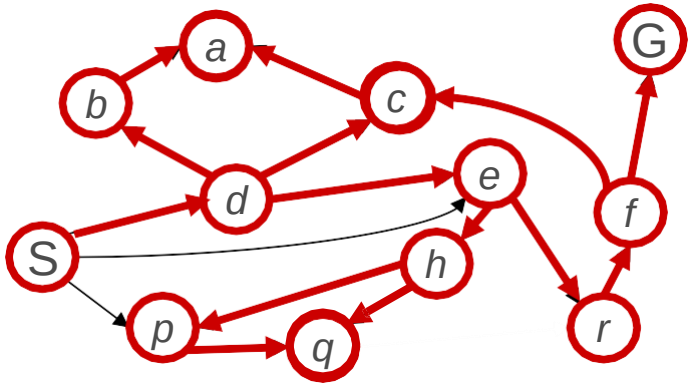
Space Complexity

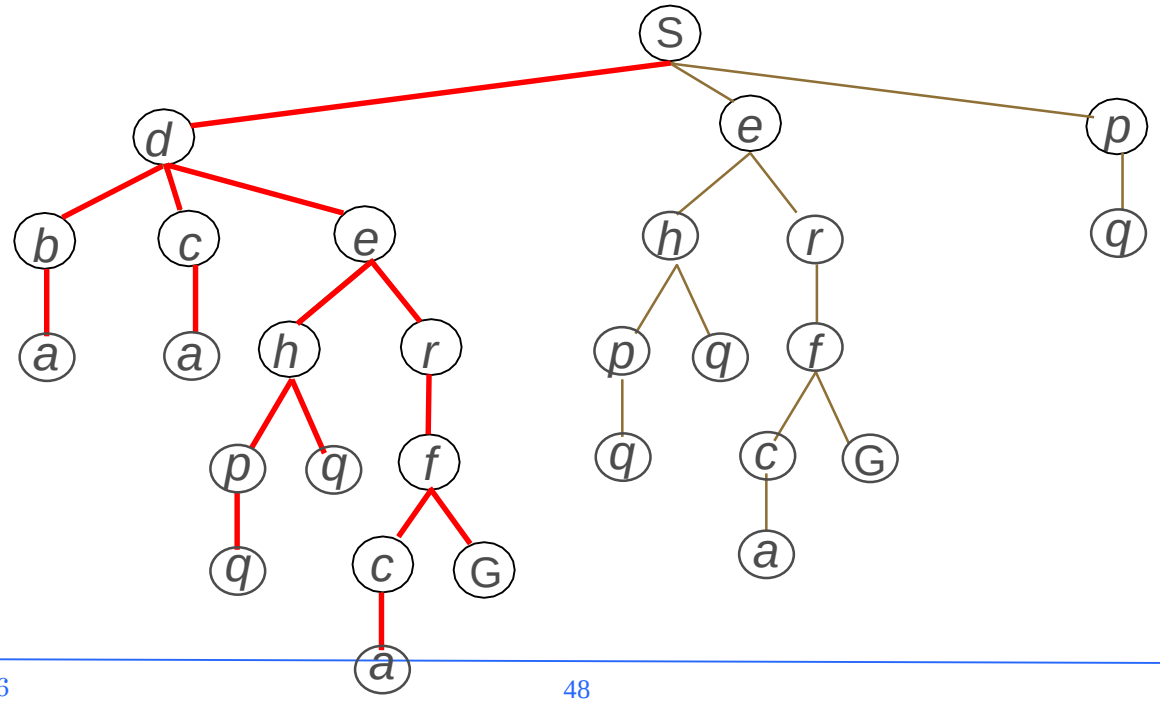
- Once a node has been expanded, it can be removed from memory as soon as all its descendants have been fully explored.
- For a state space with branching factor b and maximum depth m, depth-first search requires storage of only $O(bm)$ nodes.

Depth-First Search (DFS)

Strategy: expand a deepest node first

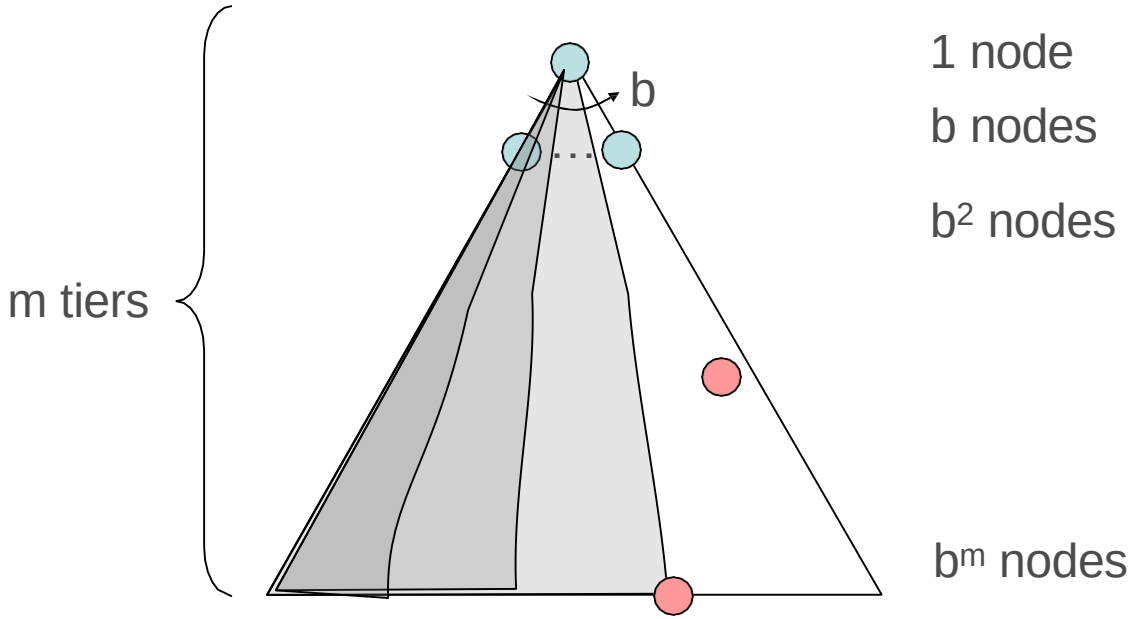
Implementation: Fringe is a LIFO stack





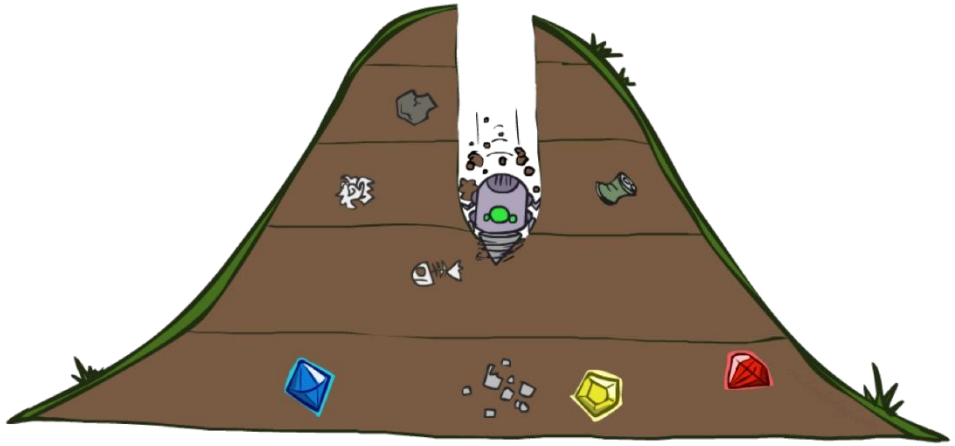
Depth-First Search (DFS) Properties

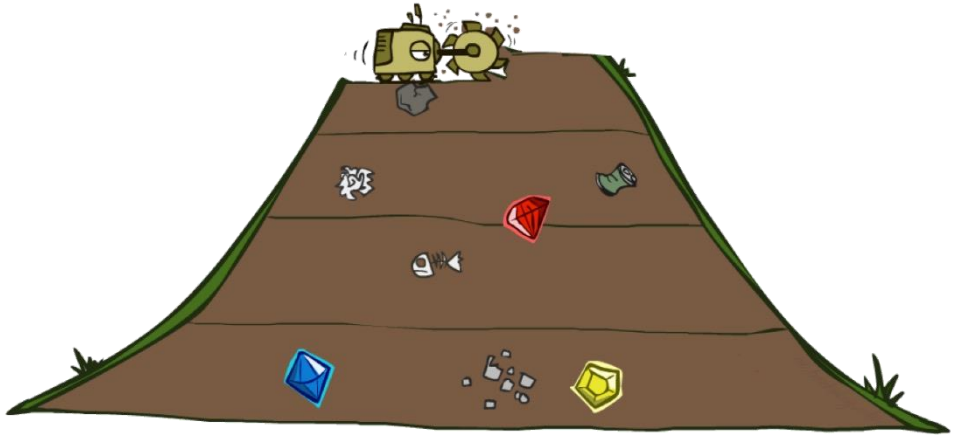
- What nodes does DFS expand?
  - Some left prefix of the tree.
  - Could process the whole tree!
  - If *m* is finite, takes time $O(b^m)$
- How much space does the fringe take?
  - Only has siblings on path to root, so $O(bm)$
- Is it complete?
  - m could be infinite, so only if we prevent cycles (more later)
- Is it optimal?
  - No, it finds the “leftmost” solution, regardless of depth or cost



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Quiz: DFS vs BFS





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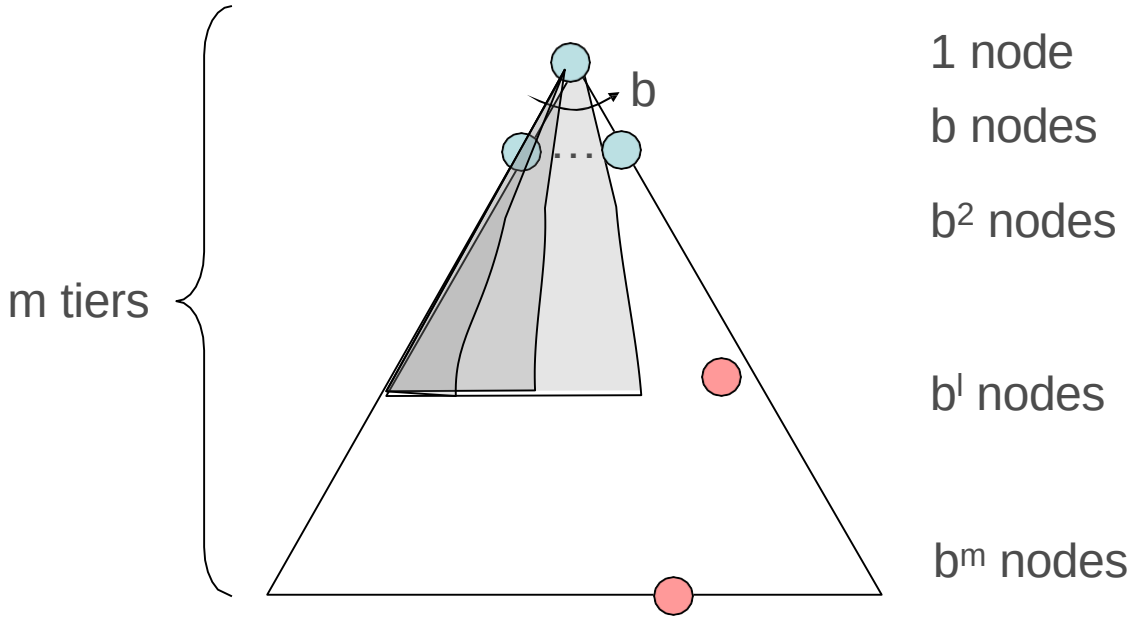
Quiz: DFS vs BFS

- When will BFS outperform DFS?

- When will DFS outperform BFS?

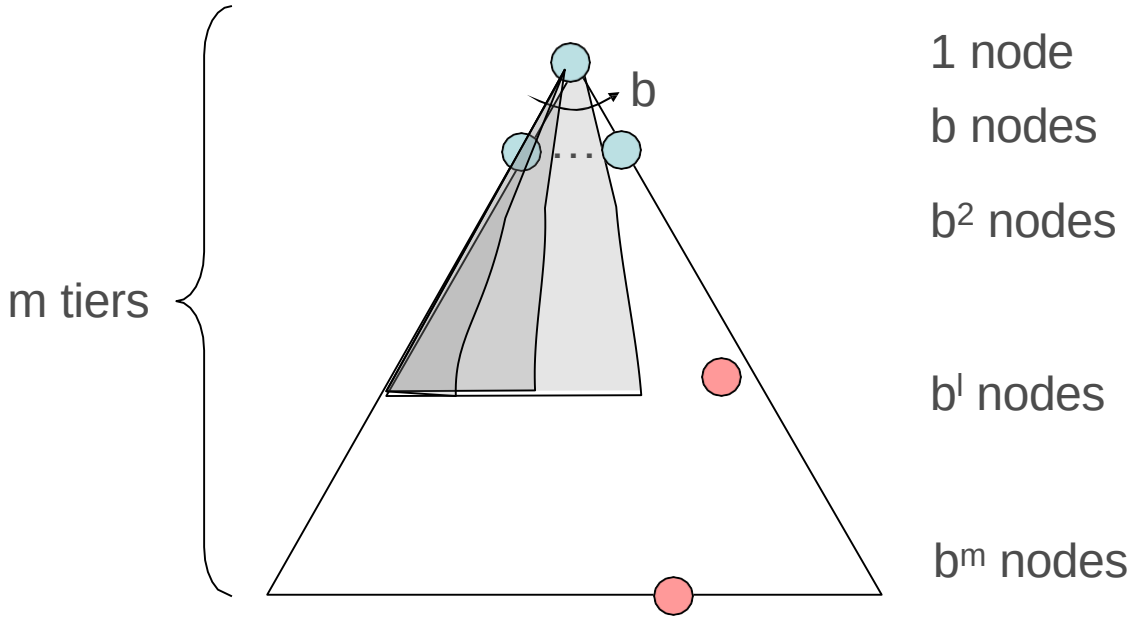
Depth-Limited Search (DLS)

- Depth Limited Search (DLS) is used for searching within a problem space by limiting how deep the search can go.
- It is based on Depth First Search (DFS) but improves efficiency by restricting unnecessary deep exploration.
- It can alliviate the failure of DFS in infinite state spaces.
- A predefined maximum depth (l) that controls how far the search can expand in the tree or graph. DFS is a special DLS with $l = \infty$.



Depth-Limited Search (DLS)

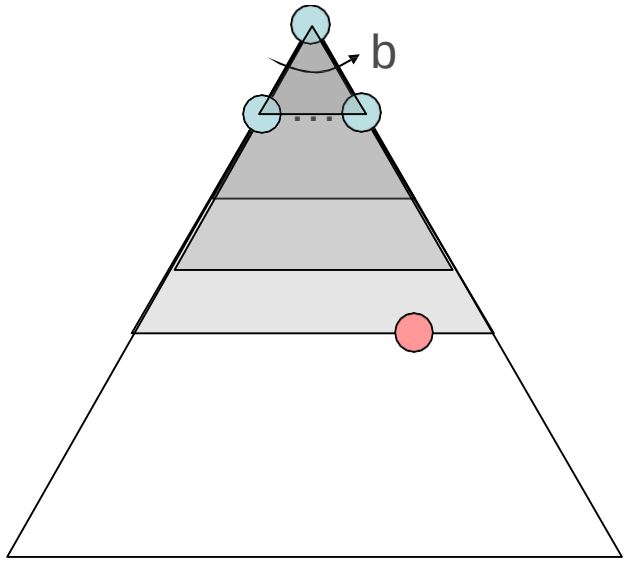
- What nodes does DFS expand?
  - Some left prefix of the tree.
  - If l is finite, takes time $O(b^l)$
- How much space does the fringe take?
  - Only has siblings on path to root, so $O(bl)$
- Is it complete?
  - only if $l \geq d$.
- Is it optimal?
  - Nonoptimal if $l > d$.



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Iterative Deepening Search (IDS)

- Idea: get DFS's space advantage with BFS's time / shallow-solution advantages
  - Run a DFS with depth limit 1. If no solution…
  - Run a DFS with depth limit 2. If no solution…
  - Run a DFS with depth limit 3. …..
- Isn't that wastefully redundant?
  - Generally most work happens in the lowest level searched, so not so bad!



- $N(\text{IDS}) = (d)b + (d - 1)b^2 + \ldots\ldots\ldots + (1)b^d$
- $N(\text{IDS}) = 50 + 400 + 3{,}000 + 20{,}000 + 100{,}000 = 123{,}450$
- $N(\text{BFS}) = 10 + 100 + 1{,}000 + 10{,}000 + 100{,}000 = 111{,}110$

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Comparing Uniformed Search Methods

Criteria	Bredth-First Search	Uniform Cost Search	Depth-First Search	Depth-Limited Search	Iterative Deepening Search
Complete?	Yes	Yes	No	No	Yes
Optimal?	Yes	Yes	No	No	Yes
Time complexity	$O(b^{d})$	$O(b^{1+[C^*/\varepsilon]})$	$O(b^{m})$	$O(b^{l})$	$O(b^{d})$
Sace complexity	$O(b^{d})$	$O(b^{1+[C^*/\varepsilon]})$	$O(bm)$	$O(bl)$	$O(bd)$



Thank you for your attention!

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